Learning and the Efficiency of Expert Referrals*

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Abstract

Buyers of complex goods and services often rely on the advice of expert intermediaries who are themselves imperfectly informed about the quality of the available options. How well do these intermediaries learn about that quality and act on it? We study this question in the setting of referrals from primary care physicians (PCPs) to specialists, using data on 4.5 million joint replacement surgeries for Medicare beneficiaries. We first document substantial heterogeneity in specialist quality and costs within geographic markets, and we present design-based evidence showing that PCPs adjust their referrals specifically based on the outcomes of their own patients. We then employ a structural learning model of PCP referral choices to quantify the losses from informational frictions and to simulate possible reallocations with improved information. Beyond learning, the model also accounts for limitations on possible reallocations due to habit persistence and capacity constraints. We find that about one-quarter of patients would be referred to a different specialist in the absence of informational frictions, with small but meaningful improvements in patient outcomes.

Keywords: informational frictions; learning; expert referrals; health care markets

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1 Introduction

Buyers of complex goods and services often cannot judge quality for themselves and instead rely on the advice of expert intermediaries. For example, investors delegate fund selection to financial advisors (Gennaioli, Shleifer and Vishny, 2015; Linnainmaa, Melzer and Previtero, 2021), employers turn to staffing agencies to screen prospective workers (Stanton and Thomas, 2016), and home buyers and sellers rely on real estate agents (Levitt and Syverson, 2008). Yet these intermediaries are themselves imperfectly informed about the quality of the available options, and in many settings, they must learn about that quality over time from the outcomes they observe. Such frictions may contribute to persistent within-market quality variation that price differences do not rationalize.[1] A natural question is how well intermediaries learn about quality differences across the options available to them, and to what extent these informational frictions shape market allocations and outcomes. We examine these questions in the setting of referrals from primary care physicians (PCPs) to specialists, a leading case in which an expert repeatedly directs others to providers whose quality is difficult to observe and revealed only gradually through experience.

A substantial literature uses structural learning models to measure how incomplete information about quality affects consumer choices and market efficiency in various industries (see Ching, Erdem and Keane, 2013, 2017, for surveys). Early work studied consumer learning about experience goods (Erdem and Keane, 1996; Ackerberg, 2003), with later applications to services such as insurance (Israel, 2005). In health care, these models are often applied to prescribing decisions, where physicians learn about the quality of new pharmaceutical drugs (e.g., Coscelli and Shum, 2004; Ching, 2010; Ferreyra and Kosenok, 2011; Chan, Narasimhan and Xie, 2013) or patient-specific matches with different drugs (e.g., Crawford and Shum, 2005; Dickstein, 2018).[2] Broadly, these papers find that uncertainty about quality substantially delays the adoption of new treatment options and limits market efficiency.[3]

In this paper, we use a structural learning model to study referrals from PCPs to orthopedic surgeons for major joint replacements. In the context of learning models, this is a novel setting

[1] See, e.g., Kolstad and Chernew (2009), Dranove and Jin (2010), and Dranove and Ody (2019) for surveys.

[2] In addition, Chernew, Gowrisankaran and Scanlon (2008) and Gong (2018) have applied structural learning models to choices about health insurance plans and surgical procedures, respectively.

[3] Crawford and Shum (2005), however, find that learning about patient-specific matches is quite fast in their application.

in which learning is done by the intermediary rather than the final consumer, and in which our data include both the signal the intermediary receives (the outcomes of each patient a PCP refers) as well as subsequent decisions based on those signals (future referrals). Referrals in general are a central mechanism in the allocation of health care goods and services in the United States, with several recent studies showing that PCP referrals greatly influence patients' choices of specialized providers (Freedman, Kouri, West and Keating, 2015; Gaynor, Propper and Seiler, 2016; Barkowski, 2018; Chernew, Cooper, Hallock and Scott Morton, 2021). Indeed, as many as 30% of visits to PCPs result in a referral to a specialist for additional care (Wright et al., 2022), and referral rates have been increasing over time (Barnett et al., 2012b; Mehrotra et al., 2011). In addition, prior research has found that referring physicians actively consider past patient experiences in future referral decisions (Schneider and Epstein, 1996; Barnett, Keating, Christakis, O'Malley and Landon, 2012a), which motivates our use of a learning model based on patient outcomes.

We estimate the model using data on over 4.5 million joint replacement surgeries in the United States from the universe of Medicare fee-for-service (FFS) claims over the years 2008 to 2018. In the model, a specialist's quality is defined as the probability of a positive outcome from the surgery. PCPs refer their patients among the set of available surgeons in their geographic market, and they learn about the qualities of those specialists from the outcomes experienced by their patients. Given the FFS environment, the PCP's referral is unaffected by insurance network restrictions, allowing a referral to any relevant specialist in the patient's geographic market. Differences in the histories of patient outcomes across PCPs referring to the same specialist provide the variation needed to identify the learning process, *net of specialist fixed effects*, which are included in the model to absorb other demand factors not related to quality. We solve both a myopic and a forward-looking version of the model, the latter using a highly tractable solution method developed by Gittins (1979) and Gittins and Jones (1979). The model also accounts for habit persistence in referral choices as well as capacity constraints using an approximation proposed by Richards-Shubik, Roberts and Donohue (2022). This is crucial for counterfactuals where informational frictions are reduced, because the capacity constraints of individual surgeons may limit the market's ability to reallocate patients toward better specialists.

Before estimating the model, we describe the referral networks and the heterogeneity in

quality across specialists within geographic markets, and we provide a simple descriptive analysis that suggests meaningful gains might be possible if patients could be reallocated. Comparing the rates of negative health outcomes (primarily surgical complications and hospital readmissions, but even death in some cases) between surgeons at the 25th and 75th percentiles within each market, we find differences of 3.97 to 6.43 per 100 surgeries in most markets. These are large differences relative to the national average failure rate of 9 per 100. We also show descriptive evidence suggesting that surgeons in the top quarter of the quality distribution may have capacity available to absorb patients otherwise treated by surgeons in the bottom quarter.

Beyond this descriptive evidence, we also examine whether PCPs adjust their referrals in response to their own patients' outcomes, a key behavioral premise of our learning model. We provide direct evidence of this behavior by comparing what a PCP does after referring a patient whose surgery ends in a failure with what the same PCP does after a referral that ends in a success. Relative to a success, a failure changes the probability that the PCP's next referral goes to the same specialist by -6.8 percent, and the relative response is largest for PCPs with little prior experience with that specialist, as Bayesian updating implies. This response is evidence of learning from the outcomes of a PCP's own patients, and it is precisely the variation that identifies the learning process in our structural model.

Turning to the results from the structural model, we find that informational frictions have substantial effects on the allocation of referrals. With perfect information about specialist quality, our counterfactual simulations show that about one-quarter of patients would be referred to different specialists, with small but meaningful improvements in health outcomes. The gains from full information are amplified by existing referral relationships, which themselves encode quality information accumulated through past referrals. When this familiarity based on past referrals is instead re-established or removed entirely, reallocation increases but the quality signal alone is too weak to direct it, with generally negative health effects.[4] We show that this pattern arises because accumulated referral relationships serve as a form of quality sorting that complements the direct effect of quality information.

Finally, as a thought experiment, we consider a simulation that imposes the quality response

[4]Reallocation with little corresponding change in patient outcomes appears in other referral settings as well. Cutler et al. (2026) find that financially motivated self-referrals from hospitals to their owned skilled nursing facilities have an imprecisely estimated zero effect on patient outcomes and Medicare spending, and Lin et al. (2025) find that hospital acquisitions of physician practices shift care toward the acquiring hospital with no improvement in patient outcomes.

to be equivalent to the effect of distance, which is the dominant non-quality factor in patients' specialist choices. This envisions a setting in which PCPs are more heavily encouraged to rely on specialist quality in making their referrals (e.g., through an EHR dashboard), while accommodating realistic constraints due to distance and specialist capacity but without any other behavioral or organizational frictions. In this simulation, which we view as an idealized upper bound, roughly half of all patients are reallocated to different specialists, and negative outcomes are reduced by nearly 20 percent.

In terms of the modeling framework in a health application, our approach is most related to working papers by Gong (2018) and Dickstein (2018), which use the same class of Bayesian learning models, also with beta distribution beliefs and binary outcomes, albeit in different contexts. Gong (2018) examines learning about a new surgical procedure for brain aneurysms, where individual surgeons learn about their own skill with the new procedure compared to an established alternative. Patient health outcomes are observed in that application, as in ours, while the physician's choice set of which procedure to pursue is much smaller than in our setting.[5] Dickstein (2018) considers antidepressant prescribing, which includes a relatively large choice set of nineteen drugs; however, health outcomes are not observed in his application, so the estimation procedure integrates out the possible sequences of outcomes and relies instead on the observed sequences of choices.

Other papers applying structural learning models to treatment decisions use a different class of models, typically assuming normally distributed beliefs and outcomes, without directly observing patient outcomes. The majority of this literature focuses on learning about the quality of prescription drugs. For example, Coscelli and Shum (2004), Crawford and Shum (2005), and Ferreyra and Kosenok (2011) study physician learning in the anti-ulcer drug market. Ching (2010), Chan et al. (2013), and Dickstein (2018) similarly consider learning with regard to side effects and treatment effectiveness of other pharmaceuticals. In addition to this work on treatment decisions, Chernew, Gowrisankaran and Scanlon (2008) study consumer learning about health insurance plan quality. Their application considers a two-period model where quality signals are observed, and they employ both beta-binomial and normal-normal models.

Beyond health care, our analysis connects to a broader literature on intermediaries who guide

[5]There are three treatment options in Gong (2018) vs. an average of around 70 orthopedic surgeons for a given HRR in our analysis.

clients under imperfect information about quality, including financial advisors (Gennaioli et al., 2015; Egan et al., 2019) and settings where relationships and reputation substitute for missing quality information when contracts are incomplete (Banerjee and Duflo, 2000; Macchiavello and Morjaria, 2015). Our setting shares the central problem in this work, in which an intermediary allocates clients across providers whose quality is imperfectly known and learned only through repeated interaction. We contribute by estimating that learning process directly from realized outcomes and quantifying its allocative consequences.

Our work also relates to the rapidly growing literature on physician networks and referrals. Starting with Barnett et al. (2011), researchers have inferred professional relationships among physicians from shared patients and have examined aggregate associations between features of the inferred networks and variations in expenditures and utilization across geographic markets. Kaur, Perloff, Tompkins and Bishop (2017) and Agha, Ericson, Geissler and Rebitzer (2022) study referrals at the provider level for specific areas of care and find that greater concentration of referrals to fewer specialists is associated with somewhat lower expenditures and utilization. Zeltzer (2020) and Sarsons (2023) further consider the role of gender homophily in referrals and the subsequent effect on earnings differences among physicians. Our descriptive analysis of the referral networks for major joint replacements offers new information on heterogeneities in network structure and quality outcomes in orthopedics, which is a prominent and growing clinical area for the Medicare population and one in which referrals can be reasonably well inferred.[6]

The remainder of the paper is organized as follows. We first explain the data and present a descriptive analysis of the referral networks for major joint replacements in Section 2, which includes a simple exercise to illustrate the potential gains from more efficient referrals within geographic markets. In Section 3, we present design-based evidence on how PCPs change their referral patterns in response to negative patient outcomes, using an event study. Section 4 turns to our learning model, where we specify the Bayesian learning process and other key features, briefly explain the solution method, and discuss identification. Finally, Section 5 describes the details of our structural estimation, provides our estimates of the myopic and forward-looking models, and presents the results from our counterfactual simulations. Section 6 concludes.

[6]Several important recent papers also consider major joint replacements (e.g., Einav, Finkelstein, Ji and Mahoney, 2020, 2022), but this literature focuses on the effects of Medicare payment reform and does not examine referrals from PCPs to specialists.

2 Data and Descriptive Statistics

To examine a predictable care pathway where patients are likely to rely on PCP referrals, our analysis focuses on planned and elective major joint replacements performed in an inpatient setting.[7] We have data covering all Medicare FFS beneficiaries aged 65 and above, from 2008 to 2018. The relevant procedures, outcomes, and referrals are identified using 100% Medicare FFS claims files, containing all Part A and Part B claims. Specifically, we identify major joint replacement procedures based on the diagnosis related group (DRG) codes on the claims, and we identify planned and elective procedures from the admission source codes also on the claims.[8] In addition to claims, we obtain data on patient characteristics from the Medicare Beneficiary Summary Files (MBSF) and data on physician practice characteristics from the Medicare Data on Provider Practice and Specialty (MD-PPAS). We also obtain data on hospital characteristics from the American Hospital Association (AHA) Annual Surveys as well as the Provider of Services (POS) files. In total, we observe just over 4.53 million inpatient stays from 2008 to 2018. Among this set, we limit our analysis to surgeries performed by an orthopedic surgeon, as indicated by the specialty codes and descriptions in the MD-PPAS data.

We infer the referring PCP based on the greatest frequency of "evaluation and management" (E&M) visits to PCPs over the prior 12-month period before a given surgery, following Pham et al. (2009) and Agha et al. (2017).[9] Eligible PCPs are indicated by the specialty codes in the MD-PPAS data, and we require that patients visited the referring PCP at least 2 times in the prior 12-month period. Restricting to procedures performed by orthopedic surgeons and where a referring PCP can be identified reduces the sample to about 2.9 million inpatient stays.

[7] During our 2008-2018 time period, major joint replacements were performed almost exclusively in an inpatient setting. CMS removed total knee replacement from its inpatient-only list at the end of our window (2018) and total hip replacement shortly after (2020), with the larger shift to outpatient and ambulatory surgical settings occurring thereafter.

[8] We consider a set of DRG codes capturing primary major joint replacement of the lower extremity (codes 469 and 470) and bilateral or multiple lower-extremity major joint procedures (codes 461 and 462). We also include anatomically related hip, knee, and upper-extremity procedures (codes 480–489, 507–508, and 510–512), while excluding *revisions* of prior joint replacements (codes 466–468) and arthroscopy (code 509). Together with the other sample restrictions described in this section, the resulting data consist overwhelmingly of elective lower-extremity major joint replacements, as presented in more detail in the supplemental appendix.

[9] Zeltzer (2020) and Sarsons (2023) instead identify referrals using the "Claim Referring Physician NPI Number" field in the carrier claims data. However in our data we find that this field is not populated in about 32% of the relevant claims, and when populated it most often lists the operating surgeon rather than a referring PCP, with only about 18% of claims naming a PCP. We present additional supporting analysis of our PCP assignment algorithm in the supplemental appendix.

Finally, we impose two additional restrictions on the sample in order to focus on physicians with sufficient experience in diagnosing and treating orthopedic patients. First, we restrict our sample to PCPs with at least 20 total referrals over our time period and with at least three consecutive years with one or more referrals. Second, we restrict our sample to surgeons with at least 20 operations per year. Our final analytic sample consists of just over 2 million planned and elective major joint replacements, in which the referring physician is a PCP with regular referrals for these procedures and in which the operating physician is an orthopedic surgeon with sufficient yearly volume.

As our measure of quality, we follow the CMS Comprehensive Care for Joint Replacement Model and the National Quality Forum. Broadly, our quality measure derives from mortality, readmissions, and complications within 90 days of discharge.[10] Since our quality measures capture negative outcomes, we sometimes refer to our measure of quality as "failures" (or as successes when measured as such). All failures are extracted from claims during and following the focal inpatient stay.[11] We also define measures of total spending and health care utilization around this episode of care by collecting all claims over the 90-day period starting with the admission date for the surgery. We employ the spending measures to assess changes in spending from reallocations as part of our counterfactual analysis.

Table 1 presents the basic descriptive statistics of our final sample. The unit of observation is one patient and procedure—i.e., an individual referral. For each referral, we observe the operating surgeon, the referring PCP, and the admitting hospital. As discussed in more detail in Section 5.2, our estimation process first divides the data into two periods: 1) the baseline period from 2008-2012, which provides a common time frame from which to form observed histories for PCP-specialist pairs; and 2) the estimation period from 2013-2018, for which the baseline period provides the initial values of the histories. From Table 1, we see fairly similar patients, physicians, practices, and outcomes between the baseline and estimation periods. One apparent difference is that PCP and specialist practice sizes are larger in our estimation period, consistent with broader national trends in practice consolidation. We also see a slight reduction

[10] As per the Comprehensive Care for Joint Replacement Model, some complications are only tracked during the inpatient stay or within 30 days of discharge. Additional details of these measures are presented in the supplemental appendix.

[11] Our baseline quality measure is not risk-adjusted, since the sample restrictions described above already select a planned, lower-complication population in which case-mix varies little across specialists. We examine sensitivity to patient case-mix adjustment in the supplemental appendix.

in readmissions, mortality, and complications in the estimation period compared to the baseline period.

2.1 Description of PCP Referral Networks

Using the physician's national provider identification (NPI) number, we identify 51,067 unique PCPs and 9,254 unique specialists over our estimation period (2013-2018). As summarized in Table 2, each PCP in our sample refers an average of 4.40 patients for elective orthopedic surgery per year to an average of 3.11 unique orthopedic specialists per year. PCPs therefore refer an average of roughly 1.41 patients to a given specialist in a year, conditional on sending any patients to that specialist that year.

We refer to the number of unique specialists referred to over some time period as the "network size" of a PCP. The distribution of PCP referral network sizes combining the six years from 2013 to 2018 is presented in Figure 1(a). This shows the total number of unique specialists to which a PCP refers patients over the estimation period, excluding the top 1% of PCPs with network sizes above 50 for ease of presentation. Consistent with the yearly average network sizes in Table 2, the typical PCP sends patients to relatively few specialists (around 10) over the entire estimation period. The frequency distribution also reveals a long tail, indicating the presence of a small set of PCPs with very large network sizes. For most PCPs, however, these network sizes are substantially smaller than the total number of orthopedic surgeons in the market, which is 69 per year on average.

To assess the concentration of PCP referral networks, Figure 1(b) presents the distribution of the proportion of a PCP's referrals going to their "highest-share" specialist. Here, we calculate each specialist's share of a given PCP's total referrals from 2013 to 2018 and take the maximum of these shares for each PCP. As is evident from Figure 1(b), many PCPs send 20% to 40% of their referrals to a single specialist, suggesting that the modal PCP's referral network is quite concentrated among a small number of specialists.

Table 2 also presents the average number of failures associated with a PCP's patients and the failure rates per referral. Failures are infrequent but not necessarily rare. On average, there are nearly 0.40 failures among a PCP's patients each year, such that around 9% of a PCP's referrals for major joint replacements result in some form of failure over our estimation period.

2.2 Within-market Variation in Specialists

As reflected in Table 2 and Figure 1(a), most PCPs have experience referring patients to between 3 and 5 unique specialists per year and around 10 unique specialists over our entire 2013-2018 estimation period. In comparison, we observe an average of nearly 69 orthopedic surgeons performing a major joint replacement per market per year, conditional on our sample restrictions discussed previously. So the average yearly network size of about 3.11 specialists indicates that an average PCP does not interact with the vast majority of available specialists in their market in a given year. This suggests a lack of experimentation with other specialists and therefore a potential opportunity for reallocation of patients across specialists in a market.

To illustrate the potential gains from such a reallocation, we consider the variation in failure rates and episode spending across specialists *within* each HRR. Specifically, Figure 2(a) plots the differences between the 75th and 25th percentiles of surgeon-specific failure rates within an HRR. This illustrates the hypothetical reduction in failure rates if PCPs could replace referrals to surgeons at the 75th percentile of failure rates with referrals to surgeons at the 25th percentile, within the same geographic market. The implication of Figure 2(a) is that failure rates could be reduced by between 3.97 and 6.43 percentage points in most markets via reallocation from high to low failure-rate specialists in the same market. More broadly, pooling across markets, the 75th-percentile surgeon has a failure rate of 13.04%, compared with 6.52% at the 25th percentile. We present a similar analysis of 90-day episode spending in Figure 2(b), which suggests possible savings of nearly $8,000 per episode when moving from specialists initiating relatively high-spending episodes (75th percentile of the market-specific spending distribution) to specialists initiating relatively low-spending episodes (25th percentile).

A simple descriptive analysis of specialist capacity further suggests that such a reallocation of patients may be feasible. Figure 3(a) presents a histogram of "potential excess capacity" among the highest-performing specialists per HRR and year. We calculate excess capacity as the difference between a specialist's yearly operations and that same specialist's 75th percentile of yearly operations across all years. We then limit to the best 25 percent of specialists with the lowest failure-rates, and we sum their specialist-specific excess capacities for each HRR and year. For comparison, Figure 3(b) presents the actual sums of operations by lowest-performing specialists (i.e., those in the highest quartile of the failure-rate distribution per

HRR and year), therefore summarizing the number of possible patients to be reallocated.[12] These two distributions suggest, descriptively, that there exists excess capacity among the highest-performing specialists in many markets, and in 39% of market-years, there appears to be sufficient capacity among top-quartile specialists to accommodate the potential reallocation from bottom-quartile specialists. While there are many other demands on specialists' time outside of surgical procedures, Figure 3 is at least consistent with the presence of excess surgical capacity to accommodate some within-market reallocations of patients.

3 Do PCPs Respond to Bad Outcomes?

Before presenting our structural learning model, we first consider design-based evidence of PCP responses to negative surgical outcomes of their patients. We estimate this response by examining how PCPs refer to specialists following good or bad outcomes from a given referral. Consider a referral of patient t from PCP i to specialist j. We define the outcome as an indicator for whether the PCP's next referral goes to the same specialist j. Treated referrals are those that resulted in a failure, and control referrals are those with no such failure. Both groups therefore consist of PCPs who have just referred a patient to specialist j.

The sample and data for this analysis are constructed as follows:

1. We begin with all referrals in our sample, weakening the restriction on specialist volume from requiring 20 operations performed in each year to requiring 20 operations performed in at least one year over our panel. We order each PCP's referrals by admission date.

2. Each referral from 2013 onward is an event. As in our model below, patients arrive sequentially, so we index both patients and time by t. For the referral of patient t from PCP i to specialist j, we denote the year by $y(t)$, and Y_{ijt} is the binary health outcome, with $Y_{ijt} = 1$ for success and $Y_{ijt} = 0$ for failure within 90 days of surgery, as in our model below.

3. For each event, we identify the PCP's next referred patient, t'. We define $R_{ijt'} = 1$ if

[12] In measuring capacity, we do not restrict to planned and elective procedures, and instead consider a broader set of all procedures in which a given specialist is listed as the operating physician. The purpose is to measure total specialist volume rather than a potentially misleading measure constructed only from planned and elective procedures.

patient t' is referred to the same specialist j, and $R_{ijt'} = 0$ otherwise.

4. We record the number of the PCP's prior referrals to specialist j before patient t, e_{ijt}, which measures the pair's referral history at the time of the event and matches the familiarity measure in our model.

With this notation, we estimate by OLS

$$R_{ijt'} = \gamma_{j,y(t)} + \beta(1 - Y_{ijt}) + \sum_{h=1}^{5} \kappa_h \mathbf{1}\left(\min(e_{ijt}, 5) = h\right) + \varepsilon_{ijt}, \tag{1}$$

where $\gamma_{j,y}$ denotes specialist-by-year fixed effects and the κ_h absorb differences in re-referral rates by the pair's referral history. Equation (1) is therefore a linear probability model, and the coefficient β captures the difference in the probability of returning to the same specialist after a failure relative to a success. We cluster standard errors by specialist and by PCP.

Table 3 presents the results. Across 1,190,521 events, of which 112,871 are failures, the estimated effect of a failure on the probability that the PCP's next referral goes to the same specialist is -0.0110 (standard error 0.0011), relative to a probability of 0.160 following a success. This corresponds to a relative change of -6.8 percent. The estimate is essentially unchanged when we add PCP fixed effects (-0.0106, standard error 0.0011).

The response also softens with the pair's prior history, consistent with standard Bayesian updating. Specifically, when the event is the PCP's first referral to the specialist, the estimated effect of a failure is -0.0126 versus a base of 0.059, a relative change of -21.2 percent. With one to four prior referrals, the relative change is -7.6 percent, and with five or more prior referrals it is -3.3 percent.[13]

To examine the timing of this response, we further estimate a stacked event study in which time is measured by the PCP's sequence of referrals rather than by calendar time. Every referral defines a window, consisting of the referral itself, the PCP's five referrals before it, and the PCP's five referrals after it, indexed by $t = -5, \ldots, 5$ with the defining referral at $t = 0$. The specialist j of the window is the one who received the referral at $t = 0$, and the outcome at each other position is $R_{ijt} = 1$ if that referral also went to specialist j. A window is a failure

[13]The relative change for pairs with five or more prior referrals is similar in magnitude to the relative reduction of 2.67 percent implied by our myopic model for pairs with non-zero referral histories, although the two quantities are defined over different samples and outcomes.

window if the surgery at $t = 0$ was a failure, $Y_{ij0} = 0$, and a success window if it was a success, $Y_{ij0} = 1$, so the comparison group again consists of PCPs who have just referred a patient to specialist j. Because every referral defines a window, the windows of one PCP overlap, and each referral appears in the windows of its neighbors as well as in its own. We keep a window only if it contains no other failure, so that a failure window contains one failure among successes and a success window contains only successes.[14] We then stack the windows and estimate

$$R_{ijt} = \mu_{ij} + \lambda_{jy,t} + \phi_{i,t} + \sum_{k \neq -1} \delta_k \mathbf{1}(t = k)(1 - Y_{ij0}) + \varepsilon_{ijt}, \tag{2}$$

where μ_{ij} is a fixed effect for each window, $\lambda_{jy,t}$ are specialist-by-year fixed effects interacted with event time, and $\phi_{i,t}$ are PCP fixed effects interacted with event time. The coefficients δ_k therefore compare failure and success windows at each event time k, relative to $k = -1$, among windows for the same specialist and year and for the same PCP. Because the PCP fixed effects are interacted with event time, δ_k is identified from PCPs with at least one failure window and at least one success window. We cluster standard errors by specialist and by PCP.

Figure 4 plots the estimates of δ_k. The requirement that a window contain no other failure reduces the sample. Of the 1,190,521 referrals in Table 3, 466,954 define a window that is kept, and 37,311 of these are failure windows. Before the event, the coefficients are close to zero, averaging -0.0020 (standard error 0.0025). At the first referral after a failure, the estimated difference is -0.0115 (standard error 0.0033), relative to a pre-event base of 0.153 among success windows. The difference persists through the third referral after the event and fades by the fifth, averaging -0.0094 (standard error 0.0026) across the five post-event referrals. Dropping as well the windows that overlap the post-period of another failure's window leaves 279,867 windows, of which 17,562 are failures. This restriction gives a difference of -0.0146 (standard error 0.0050) at the first referral after a failure, with the same pattern over the following referrals. The estimates are slightly larger when these windows are dropped, so the remaining dependence between windows, if anything, biases the main estimates toward zero. The results therefore indicate that PCPs respond to the outcomes of their own patients by redirecting subsequent

[14] A window that contains no other failure can still overlap the post-period of another failure's window, in which case the pre-period event study coefficients are potentially affected by the failure. This would introduce a SUTVA violation, as part of the response to a failure informs other event windows for the same PCP. We therefore also estimate the event study after dropping those windows as well. Details of this analysis are in the supplemental appendix.

referrals away from the specialist, albeit somewhat modestly. Relative effects are also largest when the PCP has less prior experience with the specialist.

4 Bayesian Learning Model

Our structural model describes a PCP who refers patients to orthopedic surgeons for major joint replacements. When a patient arrives who needs this treatment, the PCP recommends a specialist from the set of specialists in their geographic area. The patient follows the recommendation and receives treatment from that surgeon.[15] The outcome of the surgery is binary (success or failure), and is observed by the PCP. The probability of success, which is the definition of "quality" in this model, varies across surgeons.[16] The PCP does not know these probabilities, but they have beliefs about specialist quality that are updated based on the outcomes experienced by their patients. We solve and estimate the model for both myopic and forward-looking behavior.

In addition to the learning process, the model includes two other important mechanisms that could limit the allocation of patients to higher quality surgeons: 1) habit persistence; and 2) capacity constraints. The former is captured with a *familiarity effect* relating to the number of patients referred to a surgeon in the past, and the latter is approximated with a *congestion effect* arising from the total number of patients being treated by the surgeon at the time (i.e., referrals from other PCPs).

4.1 Model Specification

The PCP, i, refers each patient to some specialist, j, from a set of available specialists, J_i. Patients arrive sequentially, so patients and time can both be denoted by t. The choice of specialist is given by a set of indicators, $D_{ijt}, j \in J_i$, where $D_{ijt} = 1$ if patient t is referred to

[15] While some patients may ignore their PCP's recommendation or find a specialist by themselves, Freedman et al. (2015) find that the PCP's recommendation is the most commonly cited reason from a patient in their selection of an oncologist, and similar results are documented in Barkowski (2018).

[16] Throughout, we use the raw 90-day failure rate as our measure of specialist quality. The beta-binomial framework presumes binary outcomes, so it does not naturally accommodate a continuous risk-adjusted signal. Nonetheless, reduced-form multinomial logit estimates using risk-adjusted quality suggest that results based on raw failure rates are unchanged when incorporating risk adjustment. Those estimates and further discussion are presented in the supplemental appendix.

specialist j, otherwise $D_{ijt} = 0$. The binary health outcome is Y_{ijt}, with $Y_{ijt} = 1$ for success and $Y_{ijt} = 0$ for failure as defined in Section 2.

The probability of success with specialist j is q_j. This is the same for all patients, such that our model features vertical but not horizontal differentiation.[17] The PCP values the patient's outcome, which could reflect intrinsic motivations such as altruism as well as extrinsic motivations such as malpractice liability. There are other factors, denoted by x_{ijt}, affecting the patient's net benefit from a particular specialist, such as their distance to the specialist's facility, and these factors are also considered and valued by the PCP.

Beyond the patient's outcome, the PCP may value working with specialists with whom they have some prior experience. Defining $e_{ijt} \equiv \sum_{s=1}^{t-1} D_{ijs}$ as the number of past patients referred to specialist j, this *familiarity effect* is given by $f(e_{ijt})$, where f is increasing and concave.[18] In addition, the capacity constraints of individual specialists generate a negative congestion effect among patients referred to the same specialist. Let n_{jt} denote the total number of patients being treated by specialist j around time t, which includes referrals from *other* PCPs around that time (e.g., $n_{jt} \equiv \sum_k D_{kjs},\ s \in [t-\tau, t+\tau]$ for some τ), and let z_j denote fixed attributes that affect a specialist's capacity, such as non-clinical work (e.g., administrative responsibilities or academic research). The *congestion effect* is then given by $c(n_{jt}, z_j)$, where c is decreasing in n.

The PCP's realized utility from referring patient t to specialist j is

$$U_{ijt} \equiv \alpha Y_{ijt} + u(x_{ijt}) + f(e_{ijt}) + c(n_{jt}, z_j) + \xi_j + \epsilon_{ijt}, \tag{3}$$

where α is the weight on the patient's outcome, $u(x_{ijt})$ is the value placed on the other patient-specific factors, and $f(e_{ijt})$ and $c(n_{jt}, z_j)$ are the familiarity and congestion effects described

[17]The models in Coscelli and Shum (2004), Ching (2010), Ferreyra and Kosenok (2011), and Chan et al. (2013) similarly emphasize vertical differentiation among drugs, while those in Crawford and Shum (2005) and Dickstein (2018) consider horizontal differentiation in terms of the match quality between drugs and individual patients. In our case, the vast majority (86%) of treatments in our sample are for a single DRG (470), which suggests a generally similar service is being provided across patients. See the supplemental appendix for further discussion.

[18]In learning models with continuous outcomes, our notion of familiarity could instead reflect risk aversion since utility is concave in an outcome whose variance falls with experience (e.g., Erdem and Keane, 1996; Ching, 2010); however, with a binary outcome, expected utility is linear in the probability of success, and there is no curvature by which to introduce traditional risk aversion. The familiarity effect in our model therefore does not have a risk-aversion interpretation of that kind and instead reflects a direct preference for specialists the PCP has used before.

above. The term ξ_j is a specialist fixed effect, which captures time-invariant demand factors that need not be related to surgical outcomes (e.g., office amenities, health system branding, other advertising). Finally, ϵ_{ijt} is an idiosyncratic shock that captures other choice-specific factors, which has an assumed Type I extreme value distribution.

The presence of $c(n_{jt}, z_j)$ in U_{ijt} can be interpreted as the PCP internalizing the effects of congestion on the patient's net benefit from a surgeon. This can be given a structural interpretation based on waiting times, or it can be considered an approximation for other mechanisms. Either way, including this term is important to prevent bias in estimates of the responsiveness to patient outcomes, given here by α and the learning parameters, and counterfactuals based on those estimates (see Richards-Shubik et al., 2022).

4.2 Learning Process

The PCP does not know the quality of each specialist, but rather has beliefs about the possible values of q_j, for each j. These beliefs are specified with the beta distribution, denoted Beta(a, b), which is a natural and tractable modeling choice when outcomes are binary or binomial. All PCPs have the same initial beliefs (i.e., priors) about all specialists in their market, with the parameters equal to (a_0, b_0).

PCPs use Bayesian inference to update their beliefs about q_j and thereby learn about the quality of each specialist based on the outcomes experienced by their patients. Updated beliefs are governed by the numbers of successes and failures among the patients referred to specialist j in the past, as follows:

$$a_{ijt} = a_0 + \sum_{s=1}^{t-1} Y_{ijs} \quad \text{and} \quad b_{ijt} = b_0 + \sum_{s=1}^{t-1} (D_{ijs} - Y_{ijs}).$$

From the beta distribution, the mean and variance of the beliefs about q_j are given by

$$m_{ijt} \equiv \frac{a_{ijt}}{a_{ijt} + b_{ijt}} \text{ and } v_{ijt} \equiv \frac{a_{ijt} b_{ijt}}{(a_{ijt} + b_{ijt})^2 (a_{ijt} + b_{ijt} + 1)}. \tag{4}$$

Thus, m_{ijt} denotes the Bayesian expectation of the probability of success for referral t to specialist j, which is a function of the priors (a_0, b_0) and the observed outcomes among patients previously sent to that specialist. More successes will increase a_{ijt}, thus increasing m_{ijt}; however,

the magnitude of this increase is affected by (a_0, b_0). Larger values of a_0 and b_0 decrease the responsiveness of m_{ijt} to patient outcomes, and we accordingly refer to the sum of $a_0 + b_0$ as the "strength" of the priors. Intuitively, the sum $a_0 + b_0$ is like a fictitious number of patients referred to specialist j before the first actual patient.[19]

4.3 Myopic and Forward-Looking Behavior

A myopic PCP simply chooses the specialist with the highest expected payoff for the current patient:

$$\max_{j \in J_i} \mathrm{E}\left[U_{ijt} | \dots\right] = \max_{j} \left\{\alpha m_{ijt} + f(e_{ijt}) + u(x_{ijt}) + c(n_{jt}, z_j) + \xi_j + \epsilon_{ijt}\right\}. \tag{5}$$

If PCPs are myopic, they unambiguously tend to refer to specialists with whom they have had more successes in the past, all else equal. If PCPs are forward-looking, they also value experimenting with relatively unknown specialists, where the choice of specialist for referral t involves both the utility for the current patient and the value of learning more about the quality of the specialists in the market, which could benefit future patients. The solution to this dynamic problem simplifies with the use of a *Gittins index* (Gittins, 1979; Gittins and Jones, 1979), which expresses the value of learning about specialist j as a function of the mean and variance of the current beliefs about that specialist. We denote the index abstractly as $g(m_{ijt}, v_{ijt})$, but this function is well approximated with a fairly simple and tractable closed-form expression developed by Brezzi and Lai (2002).

The Gittins index requires four key assumptions: (i) one option is chosen at a time; (ii) the unchosen options do not affect the current outcome; (iii) the expected returns do not change for the unchosen options; and (iv) beliefs are independent across options. These assumptions are not uncontroversial, but they are relatively unproblematic in our application. Typically, (i) one surgeon treats the patient, and (ii) other surgeons do not assist or interfere with the operation. For (iii) and (iv), we assume that PCPs perceive surgeons to have fixed quality that is independent across surgeons. There would be no depreciation of skills among unchosen surgeons (i.e., unchosen surgeons continue to receive patients from other PCPs) and no important

[19] Because the variance decreases in $a + b$, this sum is sometimes referred to as the "precision" of the beliefs (analogously, $a_0 + b_0$ is the precision of the priors).

spillovers in quality among surgeons. To this last point, we may rely on the surgeon fixed effects to absorb time-invariant correlations among surgeons, for example among those who operate in the same hospitals.

With forward-looking behavior, the PCP's present discounted utility is given by value functions, which can be written abstractly as follows:

$$V_{it}(\dots) = \max_{j \in J_i} \left\{ \mathrm{E}\left[U_{ijt} | \dots\right] + \beta \mathrm{E} V_{i,t+1}(\dots) \right\},$$

where β is a known discount factor, and the ellipses $(\dots)$ represent the relevant state variables. These value functions simplify because there are no dynamics in x, n, z, ξ, and ϵ. Patients are assumed to arrive randomly, which makes x exogenous, z and ξ are fixed over time while ϵ is independent, and we assume there are enough patients and PCPs such that no individual PCP can affect n. Hence the components of the future value function related to these terms do not vary across choices in the current period. Because they are also additively separable in the utility function, these components of the future value function drop out of the differences across choices, and therefore they can be ignored in estimation. This leaves the value of learning about the quality of a specialist, captured by the Gittins index, and the present discounted value of increasing familiarity with a specialist. The former has a closed-form approximation as noted above, and the latter is computed as the present discounted value of sending every future patient to the same specialist, denoted by $\overline{\overline{f}}(e_{ijt})$.[20]

Thus, the forward-looking choice problem simplifies as

$$\max_{j \in J_i} \left\{ \alpha g(m_{ijt}, v_{ijt}) + \overline{\overline{f}}(e_{ijt}) + u(x_{ijt}) + c(n_{jt}, z_j) + \xi_j + \epsilon_{ijt} \right\}. \tag{6}$$

Unlike the myopic model, the forward-looking model assigns some value to experimenting with specialists with whom the PCP has less past experience. That is because the Gittins index is increasing in the variance, v_{ijt}, and the variance is decreasing in the number of past patients (as seen in Equation (4), v_{ijt} is decreasing in $a_{ijt} + b_{ijt}$). It is therefore possible for the Gittins index to assign a higher value to a specialist with a lower current expectation for the probability of

[20]The Gittins index is defined as a lump sum payment that makes an individual indifferent between using one option and retiring it. Because f is increasing in e, the value from f is maximized by sending every future patient to the currently chosen specialist.

success but a larger variance: e.g., $m_{ijt} < m_{ikt}$, $v_{ijt} > v_{ikt}$, and $g(m_{ijt}, v_{ijt}) > g(m_{ikt}, v_{ikt})$.

4.4 Identification

Here we discuss the identification of the parameters in the myopic model and offer a simple conjecture for the forward-looking model. We also discuss simulation results that indicate the difficulty of estimating the strength of the priors $(a_0 + b_0)$ in practice.

First, given a distribution for the shocks ϵ, the conditional choice probabilities identify differences in utility between the alternatives. The effects of the pairwise characteristics, $u(x)$, and the specialist fixed effects, ξ, are identified subject to standard normalizations (e.g., one fixed effect is set equal to zero). Because the number of patients at a specialist, n_{jt}, is endogenous, the congestion effect, $c(n_{jt}, z_j)$, is identified with an instrument for the probability that patients *other than patient t* see specialist j. A natural instrument for this is the distances between *other* patients and specialist j (see the description of the estimation procedure in Section 5.3, and Richards-Shubik et al., 2022, for further details).

This leaves the terms αm and $f(e)$ in Equation (5). The parameter α is identified in the limit as the number of past patients sent to a specialist grows large (i.e., so-called "identification at infinity").[21] To show this, recall $e_{ijt} \equiv \sum_{s=1}^{t-1} D_{ijs}$ and let $\bar{y}_{ijt} \equiv \sum_{s=1}^{t-1} Y_{ijs}/e_{ijt}$. Then αm_{ijt} from Equation (4) can be rearranged as

$$\alpha m_{ijt} = \alpha \frac{a_0}{(a_0 + b_0) + e_{ijt}} + \alpha \frac{e_{ijt}}{(a_0 + b_0) + e_{ijt}} \times \bar{y}_{ijt}. \tag{7}$$

In the limit as $e_{ijt} \to \infty$, we have $\alpha m_{ijt} = \alpha \cdot 0 + \alpha \cdot 1 \cdot \bar{y}_{ijt}$. Thus, intuitively, α is identified by variation in the success rate among specialists to whom a PCP has sent many patients in the past.

For the next step in the argument, it is useful to re-parameterize a_0 and b_0 as $\eta \equiv (a_0 + b_0)$ and $\rho \equiv \frac{a_0}{a_0 + b_0}$. The strength of the priors, η, is identified by the interaction between e_{ijt} and $\bar{y}_{ijt}$ when e_{ijt} is finite. This interaction appears only in the second term in (7), where η is the remaining unknown parameter (since α is identified in the limit). Because η is positive, the marginal effect of the success rate ($\bar{y}_{ijt}$) increases with the number of past patients (e_{ijt});[22] η

[21] A related argument is given verbally in Ching et al. (2013), for a normal-normal learning model.

[22] Note that $e/(\eta + e)$ goes from 0 to 1 as e goes from 0 to ∞.

governs this interaction—i.e., how quickly the marginal effect of the success rate increases with the number of past patients referred to a specialist.

To identify the prior mean, ρ, we use the observed average success rate in the market. This assumes that PCPs have a general awareness of the overall success rate of the procedure in their local area, and have rational expectations.[23] It might instead be possible to identify ρ (equivalently a_0) from the choice data, but this would require a strong functional form assumption to separate the first term in (7), $\alpha\frac{a_0}{(a_0+b_0)+e}$, from the familiarity effect, $f(e)$.

Finally, then, the familiarity effect is identified by the remainder of the marginal effect of experience with a specialist that is not absorbed by αm (i.e., what is left over from $\alpha\frac{a_0}{(a_0+b_0)+e}$). A standard normalization is required, such as $f(0) = 0$. Also f must be bounded as $e \to \infty$; otherwise, familiarity would dominate the effects of all other attributes as $e \to \infty$. This does not seem desirable from a modeling perspective, and it would nullify the argument for the identification α in the limit.

The argument above clarifies the key sources of variation that identify each parameter, in particular α, along the lines of Ching et al. (2013). This parameter, which gives the utility weight on the outcome (or "use experience") is omitted from the models in Crawford and Shum (2005), Ferreyra and Kosenok (2011), Dickstein (2018), and Gong (2018) but included in those in Ching (2010) and Ching et al. (2013), for example. In some cases, this omission may be innocuous because outcomes are not observed, so the scale of the outcome is estimated and naturally incorporates the utility weight (i.e., the scale is in terms of utility). However, it is not clear that this is unrestrictive when outcomes are binary, as in Dickstein (2018) and Gong (2018), which fixes the scale of the experienced outcome even if it is unobserved. This is potentially important because the shocks in these models have a parametric distribution with an assumed scale. Consequently, fixing the utility weight on the outcome would imply that it has the identical impact on utility as all the factors captured in the shocks—or equivalently, the marginal rate of substitution between the outcome and the shocks is assumed to be known.

Separately, simulation exercises suggest that the strength of the priors, $\eta \equiv (a_0+b_0)$, may be poorly identified in practice (see the supplemental appendix for details). While α is recovered reasonably well in simulated data, η is noisy, and the likelihood function with respect to η is

[23] Coscelli and Shum (2004), Crawford and Shum (2005), Chan et al. (2013), and Dickstein (2018) similarly use a rational expectations assumption to identify the prior beliefs in their models.

relatively flat. For some intuition into the problem, consider the second term in Equation (7), which shows how η is identified from the interaction between the number of past patients (e_{ijt}) and the success rate among those patients ($\bar{y}_{ijt}$), as discussed above. From that term (omitting α), the cross partial of e and $\bar{y}$ would be $1/(\eta+e) - e/(\eta+e)^2$, which exhibits little variation over a range of values for η. For example, using the mean number of past patients (conditional on any past referrals from i to j), shown in Table 2 (see "running referrals"), this cross partial ranges from 0.033 to 0.056 when η ranges from 1 to 20, which is little variation relative to the scale of the error terms in the model. Consequently, for the estimation of the model, we fix η at one of a set of alternative values.

Last, we turn to the identification of the forward-looking model. From Brezzi and Lai (2002), the Gittins index is closely approximated by

$$\tilde{g}(m,v) \equiv m + \sqrt{v} \cdot \psi\left(\frac{v}{-\ln(\beta)\, m(1-m)}\right), \tag{8}$$

where ψ is a known function derived by Brezzi and Lai (2002), and β is a known discount factor.[24] Substituting $\tilde{g}$ for g, the forward-looking model shown in Equation (6) has no additional free parameters beyond those of the myopic model in Equation (5). The only differences between the myopic model (5) and the dynamic model (6) are that $\tilde{g}(m,v)$ and $\overline{\overline{f}}(e)$ appear in place of m and $f(e)$, respectively. The Gittins index approximation, $\tilde{g}(m,v)$, involves the same free parameters as the mean of the current beliefs, m, which are a_0 and b_0 (or equivalently, ρ and η), and the same data (the numbers of past referrals and successes). As for the present discounted value of familiarity, $\overline{\overline{f}}(e)$, it has a known relationship with the current value of familiarity (see Gong, 2018, for an equivalent result applied to an effect of learning by doing). Thus, while this remains a conjecture, it seems reasonable to suppose that the forward-looking model is identified under the same conditions as the myopic model.

5 Estimation of the Learning Model

Below we provide details of the empirical specification, variable definitions, and estimation procedure. We then present the parameter estimates and counterfactual results.

[24] Ching et al. (2013) discuss the conditions and challenges for identifying the discount factor in structural learning models. We treat it as a known parameter.

5.1 Empirical Specification

Recall the maximization problem for a myopic PCP from Equation (5),

$$\max_{j \in J_i} \mathrm{E}\left[U_{ijt} | \dots\right] = \max_{j} \left\{\alpha m_{ijt} + f(e_{ijt}) + u(x_{ijt}) + c(n_{jt}, z_j) + \xi_j + \epsilon_{ijt}\right\}.$$

In estimating this equation, we assume that the idiosyncratic shocks, ϵ, have a Type I extreme value distribution, which yields a multinomial logit model. The fixed effects, ξ, are free parameters, which are separately identified due to the large number of patients relative to the number of surgeons per market. Congestion is assumed to be constant within pre-specified time periods (e.g., a calendar year), so that $c(n_{jt}, z_j) \equiv c(n_{j,\tau(t)}, z_j)$, where $\tau(t)$ is the period containing t. This is for computational feasibility because, in the first step of estimation (described below), the congestion effect is absorbed into a composite fixed effect, $\delta_{j,\tau(t)} \equiv c(n_{j,\tau(t)}, z_j) + \xi_j$, and one of these must be recovered for each surgeon in each relevant time period. The capacity shifters, z_j, are omitted due to data limitations, and we specify the congestion effect to be linear so that $c(n_{j,\tau(t)}, z_j) \equiv \gamma n_{j,\tau(t)}$. There is one patient-specific factor, distance to the surgeon's facility, specified with a linear effect on utility: $u(x_{ijt}) \equiv \pi x_{ijt}$. The effect of familiarity, $f(e_{ijt})$, is specified non-parametrically using a series of indicators for different levels of e_{ijt}. Specifically, we separate e_{ijt} into bins [0], [1], [2], [3], [4], [5], $[6, 7]$, $[8, 10]$, $[11, 15]$, $[16, \infty)$ and include indicator variables κ_b for each bin, b, excluding κ_0.

For the prior beliefs, we use the re-parameterization introduced in Section 4.4:

$$\rho \equiv \frac{a_0}{a_0 + b_0} \quad \text{and} \quad \eta \equiv a_0 + b_0, \tag{9}$$

which are respectively the mean and strength of the priors. Hence, the mean of the current beliefs about the quality of specialist j in period t is specified as

$$m_{ijt} = \frac{\rho\eta + y_{ijt}}{\eta + e_{ijt}}, \tag{10}$$

where $y_{ijt} \equiv \sum_{s=1}^{t-1} Y_{ijs}$ is the number of success among patients referred to j in the past, and $e_{ijt} \equiv \sum_{s=1}^{t-1} D_{ijs}$ is the number of patients referred to j, as defined previously. The expected

utility for a myopic PCP is thus:

$$\mathrm{E}\left[U_{ijt}|\dots\right] = \alpha\frac{\rho\eta + y_{ijt}}{\eta + e_{ijt}} + \pi x_{ijt} + \sum_{b\in B}\kappa_b I(e_{ijt}\in b) + \underbrace{\gamma n_{j,\tau(t)} + \xi_j}_{\delta_{j,\tau(t)}} + \epsilon_{ijt}. \tag{11}$$

As discussed in Section 4.2, the analogous expected utility for a forward-looking PCP is

$$\mathrm{E}\left[U_{ijt}|\dots\right] = \alpha\tilde{g}(m_{ijt}, v_{ijt}) + \pi x_{ijt} + \sum_{b\in B}\kappa_b I(e_{ijt}\in b) + \underbrace{\gamma n_{j,\tau(t)} + \xi_j}_{\delta_{j,\tau(t)}} + \epsilon_{ijt}, \tag{12}$$

where $\tilde{g}(m_{ijt}, v_{ijt})$ is approximated using Equation (8). In both the myopic and forward-looking models, the parameters $\alpha, \pi, \kappa_1, ..., \kappa_B$, and the composite fixed effects $\delta_{j,\tau(t)}$ are estimated from the referral decisions, while ρ is set equal to the mean success rate in the market and η is given one of a set of fixed values, as discussed in Section 4.4.

5.2 Variable Definitions

For each PCP-specialist pair, the running variables e_{ijt} and y_{ijt} give the numbers of referrals and successes prior to patient and time t. However, because our sample begins in 2008, we do not observe the complete history for every pair. Experiences from many years in the past may also be less salient for current referral decisions. To address these issues and employ comparable observation periods to construct histories for all PCPs and specialists, we limit the running variables to a five-year look-back period.[25] Referrals and outcomes occurring more than five years in the past are thereby assumed not to affect the current referral decision. While this five-year period is somewhat arbitrary, some cutoff is necessary in order to ensure a common length of history for all PCP-specialist pairs. We selected five years as it seems reasonably long, and it splits our available data into two similar periods: 2008-2012 is the baseline period from which the initial histories are constructed, and 2013-2018 is the estimation period. The specialist-level patient volumes, $n_{j,\tau(t)}$, which are used for the congestion effect, are defined using three-year periods (2013-2015 and 2016-2018).

We construct choice sets based on the patient's HRR. For concreteness, denote a patient

[25]For example, consider a referral from PCP i to specialist j for an inpatient admission on January 1, 2017. Our running variables for this referral are constructed using admissions from January 1, 2012 through December 31, 2016.

t's HRR as h_t. We then take the set of specialists operating on any patient in h_t, limiting to specialists with at least 20 surgeries in a calendar year. This initial choice set might include specialists located outside of a given HRR but that nonetheless operate on patients residing in h_t. Next, we exclude any specialists whose nearest practice location is farther than the 90th percentile of distances to other specialists in that patient's choice set, or beyond 75 miles. The distance between the patient and the specialist is computed using the zip codes of the patient's residence and the specialist's nearest practice location. The resulting choice sets contain about 57 orthopedic surgeons, on average, compared to 69 specialists per HRR per year on average, so our distance restriction does not severely limit the effective choice sets.

5.3 Estimation Procedure

The specifications in Equations (11) and (12) yield relatively standard multinomial logit models, with the exception of the congestion effect. Richards-Shubik et al. (2022) provides a two-step approach to estimate these models with a congestion effect, which is based on methods from Bayer and Timmins (2007) and closely related to the demand estimation from Berry, Levinsohn and Pakes (1995).

In the first step, the congestion effect, $\gamma n_{j,\tau(t)}$, is absorbed into the composite fixed effect, $\delta_{j,\tau(t)}$ in Equations (11) and (12), and the resulting multinomial logit is estimated via maximum likelihood. In the second step, γ is estimated with a linear regression defined by the identity $\delta_{j,\tau(t)} \equiv \gamma n_{j,\tau(t)} + \xi_j$. Because $n_{j,\tau(t)}$ is endogenous to ξ_j, and since ξ_j represents part of the error term in this second step regression, we estimate γ via two-stage least squares. We employ as an instrument the predicted number of patients, $\hat{n}_{j,\tau(t)}$, generated using a simple multinomial logit with only distance as the independent variable. In relation to the individual referral decision in the first step, the exclusion restriction for this instrument comes from the distances between *all other patients* and specialist j. Specifically, $\hat{n}_{j,\tau(t)}$ is generated using the distances, x_{kjs}, from all PCPs k and patients s in the market during the relevant time period, while only the current patient's distance, x_{ijt}, affects the given referral decision.[26]

The estimation of the multinomial logit in the first step is further simplified because ρ is

[26]First-stage F-statistics comfortably exceed conventional weak-instrument thresholds in both the pooled specification and the large majority of HRR-level specifications. We present the first-stage diagnostics in the supplemental appendix.

set equal to the mean specialist quality in that HRR and η is fixed at predetermined values of 1 and 5.[27] The utility specification is therefore linear in the remaining parameters. Finally, we constrain α to be weakly positive so that successful patient outcomes ($Y_{ijt} = 1$) are weakly preferred by PCPs. We impose this constraint using an exponential transformation (i.e., $\alpha = e^{\tilde{\alpha}}$), implemented using the algorithm proposed in Fader et al. (1992). In our setup, that algorithm amounts to an iterative procedure in which we propose an initial value, $\tilde{\alpha}_0 = \ln(\alpha_0)$, construct the variable $\tilde{m}_{ijt,0} = e^{\tilde{\alpha}_0} \times m_{ijt}$, estimate the coefficient on this transformed variable, $\widehat{\Delta}_0$, and then update to $\tilde{\alpha}_1 = \tilde{\alpha}_0 + (\widehat{\Delta}_0 - 1)$. This process continues until $\tilde{\alpha}$ converges, which we denote as $\tilde{\alpha}^*$. Our final estimate for α then follows from the transformation of $\tilde{\alpha}^*$, so that $\hat{\alpha} = e^{\tilde{\alpha}^*}$.[28]

Four alternative models are estimated: assuming myopic or forward-looking behavior, and using $\eta = 1$ or $\eta = 5$. We estimate these models separately in each of the 306 HRRs in our data, both to examine variation across markets and for computational feasibility, avoiding the simultaneous estimation of many thousands of specialist FEs. The distributions of estimates across HRRs are summarized by taking a weighted average of the estimates from each HRR, weighted by the number of observations per HRR. The congestion effect, γ, is estimated separately in each HRR along with the other parameters, and it is also estimated nationwide by pooling the composite fixed effects ($\hat{\delta}_{j,\tau(t)}$) and predicted numbers of patients ($\hat{n}_{j,\tau(t)}$) across all HRRs, with HRR fixed effects included in the pooled 2SLS specification.[29]

In the counterfactuals, a change in where patients go changes each surgeon's volume and with it the congestion term. We therefore solve for the fixed point at which each surgeon's volume equals the sum of the counterfactual referral probabilities of the patients in the market, starting from the observed volumes and iterating until volumes and choices are consistent. The supplemental appendix shows that the solution is the same from two extreme starting values.

[27] From Table 2, PCPs refer an average of 4.40 patients per year in total, or about 4.44 cumulative patients to a given specialist during the estimation period, so these values for η span a reasonable range for the strength of the priors in relation to the information learned from actual patients.

[28] The supplemental appendix provides the full derivation of this iterative procedure and the implied standard error for $\hat{\alpha}$.

[29] The HRR fixed effects in the second-stage regression account for the fact that specialist fixed effects in the first step are estimated separately within each HRR, with one specialist's fixed effect omitted per market as the reference category.

5.4 Parameter Estimates

Out of 306 HRRs, our constrained maximum likelihood estimation converged in over 90% of markets, with slight variation across the myopic versus forward-looking models and the values of η.[30] Parameter estimates from the myopic model are presented in Table 4, with analogous results from the forward-looking model in Table 5. The tables present the coefficient estimates of α, π, γ, and κ using different values for η, as indicated, along with the estimates of the prior mean ρ that come from the observed success rates (and therefore do not vary with η). We summarize the estimates for familiarity focusing on the intervals up to 10 patients for brevity. As noted above, α, π, and κ are estimated separately in each HRR while γ is estimated by pooling across HRRs. For α, π, and κ, we present the mean across HRRs along with certain percentiles.

In both the myopic and forward-looking models, the estimates of π are in line with other estimates of the effect of distance found in the literature, and are fairly similar across the values of η. The estimates for familiarity are also similar across η and are generally large and increasing, suggesting that familiarity plays a potentially large role in driving referrals. Finally, the estimates of γ are negative and significant, indicating the presence of congestion effects such that specialists with larger patient volumes are somewhat less likely to receive a given referral. The estimates of about -0.1 (per 100 patients) imply that specialists with 100 more patients (over a 3-year period) are 7-9% less likely to receive a particular patient, depending on the baseline referral probability.[31]

The estimates of α indicate that PCPs place small to moderate weights on patient outcomes depending on the market. Larger values of η yield larger estimates of α, which is natural because α is divided by $\eta + e$ to form a composite "coefficient" on y in Equation (11), with a similar scaling effect in the forward-looking case. We estimate slightly larger values for α in the forward-looking models, consistent with the additional informational value from patient outcomes in a forward-looking versus myopic decision. Focusing on the smallest estimate of the mean value of α, in the myopic model with $\eta = 1$ (in Table 4, column "Mean"), its ratio to the

[30] Specifically, in the myopic (forward-looking) case, the estimation converged for 280 (277) markets for $\eta = 1$ and 281 (280) markets for $\eta = 5$.

[31] In a multinomial logit, the relative marginal effect on a choice probability p_j is $(1 - p_j)\gamma$. The average referral probability is about 0.33 for a PCP's most-chosen specialist and around 0.10 for any specialist with a past referral, yielding relative effects of -0.067 and -0.090 respectively with $\gamma = -0.1$.

corresponding mean estimate of π implies that on average PCPs view the difference between success and failure as worth about 2.91 miles of extra travel distance for the patient. Such a small marginal rate of substitution is consistent with other estimates in the literature finding small quality elasticities of demand. Also, the national mean estimate of α is reasonably precise. A simple estimate of its standard error can be computed by dividing the standard deviation across HRRs (in column "SD/SE") by the square root of the number of HRRs, which gives a naive standard error of 0.0107.[32]

To further interpret the role of patient outcomes in PCP referrals, we present the partial effect of one additional failure in Figure 5 among PCP-specialist pairs with non-zero referral histories. The figure plots the distribution of average partial effects across HRRs. It illustrates two key findings. First, there are some markets in which PCPs essentially do not respond meaningfully to specialist failures (i.e., $\hat{\alpha} = 0$). Second, in markets where there is a response, one additional failure reduces the referral probability by 0.32 percentage points, on average, which is a small but meaningful relative reduction for pairs with non-zero referral histories.[33]

5.5 Counterfactuals

Using our model of PCP learning and referrals, and the estimates summarized in Tables 4 and 5, we consider two primary counterfactuals. Both sets of counterfactuals simulate the reallocation of patients and the effects on patient health under full information, in which PCPs have perfectly accurate beliefs about specialist quality.[34] The counterfactuals differ in regard to our assumptions on the role of familiarity. In one case, we initialize familiarity as observed in the data as of the time of the first referral, and in the other, we initialize familiarity to 0. We refer to the first set of counterfactuals as "existing familiarity" and the second set as

[32]These estimates rest on the interpretation that the specialist fixed effects capture demand factors distinct from quality rather than absorbing quality itself, and that the estimator recovers α reliably in the relevant range. In the supplemental appendix, we show that the within-HRR correlation between $\hat{\xi}_j$ and observed specialist quality is near zero in most markets, and a Monte Carlo study indicates that $\hat{\alpha}$ is recovered with low relative bias over the range where our non-zero estimates fall.

[33]Our reduced-form evidence in Section 3 measures the change in the probability that a PCP's next referral goes to the same specialist, whereas the partial effect here is the change in the referral probability for a given patient, so the two are defined over different outcomes and samples. The relative reduction of 2.67 percent implied here is nonetheless similar in magnitude to the relative change for pairs with five or more prior referrals in Table 3.

[34]This is implemented by replacing m in the myopic version and g in the forward-looking version of the model with the true probability of success for each specialist, as observed in the data. For the Gittins index, note that when the variance of the beliefs, v, is zero, the expression in Equation (8) becomes just m.

"resetting familiarity."[35] In both cases, the number of patients referred to each specialist can change going forward, which naturally allows the familiarity effect to change along with quality information. Familiarity can therefore compound the effect of improved quality information, or of other characteristics if the quality response is sufficiently low.[36]

The results under "existing familiarity" are summarized in Figure 6. With existing familiarity, full information leads to a reallocation of 23.25% of referrals in the myopic model, with an average improvement in health outcomes of 0.07 per 100 patients (about a 0.8% relative reduction against the baseline failure rate of 9 per 100). Beneath this aggregate, the across-HRR distribution of mean health effects is skewed positive, with roughly three-quarters of markets showing improvements and a longer right tail of larger gains. These results include adjustments for changes in congestion, as discussed in Section 5.3, and are relatively similar across different values of η and across the myopic and forward-looking models. We examine other changes in the within-market distributions of specialist quality in the supplemental appendix, which presents the changes in the 10th and 90th percentiles of referral-weighted quality in each HRR. Beyond the changes in the mean shown in Figure 6, it appears that the reallocation under existing familiarity yields modest upward shifts at the low end of the distributions more than at the high end.

The results differ meaningfully when we reset familiarity. As summarized in Figure 7, the share of referrals reallocated increases to 31.75%, but average health effects turn slightly negative at -0.02 per 100 patients (about a 0.2% relative increase against the baseline failure rate of 9 per 100). The across-HRR distribution under resetting familiarity is wider and roughly symmetric around zero, with about half of markets showing improvements; the slightly negative patient-weighted mean reflects larger magnitudes among the worsening markets. Under this resetting-familiarity counterfactual, the quality distribution widens rather than compresses upward. A minority of patients are reallocated to higher-quality specialists they had previously been far from, while another group loses the quality cues embedded in their PCP's prior re-

[35]Under "existing familiarity," the counterfactuals simulate a shock where the PCPs unexpectedly receive perfect information about specialist quality on January 1, 2013.

[36]In the supplemental appendix, we also consider a simulation in which familiarity is shut off entirely. Removing familiarity produces even larger reallocations but negative average health effects, reinforcing the finding that existing referral relationships encode quality information that the quality signal alone cannot replace. The appendix also verifies that our counterfactual equilibrium is robust to starting values, with HRR-level mean quality differing by at most 7×10^{-5} when re-solving from two extreme initial conditions.

ferral patterns and is reallocated to lower-quality specialists. The net mean effect on quality is slightly negative because the losses at the bottom of the distribution within HRRs typically outweigh the gains at the top (see the supplemental appendix).

The comparison between the two counterfactuals reveals an important interaction between quality information and existing referral relationships. When PCPs retain their accumulated familiarity, full information nudges referrals toward higher-quality specialists largely within the PCPs' existing referral networks. When familiarity is reset, the reallocation is driven primarily by distance preferences and specialist fixed effects rather than by quality, because the estimated weight on quality (α) is small relative to the dispersion of specialist fixed effects. To illustrate, the within-market standard deviation of specialist fixed effects is on average over 7 times larger than the quality signal (α times the standard deviation of specialist quality), so that the quality channel is overwhelmed once familiarity no longer anchors referral patterns.

These patterns emerge more clearly when we condition on markets where PCPs demonstrably respond to outcomes. Figure 9 plots the mean health effect of each counterfactual against the estimated learning parameter $\hat{\alpha}$ across markets with $\hat{\alpha} > 0.001$, binned into ten patient-weighted quantiles. Under both counterfactuals, health effects scale strongly with $\hat{\alpha}$. Markets with the largest $\hat{\alpha}$ see the most meaningful health improvements when full information is combined with existing familiarity, and even under the resetting-familiarity counterfactual, health effects turn less negative (or positive) in these responsive markets.

As a separate check on the nature of these reallocations, we also examine whether patients are being moved toward specialists with higher or lower spending at baseline. We treat this as a descriptive diagnostic, not a counterfactual simulation, because we do not model spending behavior directly. Under existing familiarity, average episode spending (estimated from specialists' observed patients) falls modestly (Figure 8). This suggests that the quality-improving reallocations do not necessarily tilt toward more expensive specialists. When resetting familiarity, average spending changes are small. The co-movement between quality and spending is somewhat mechanical, however, because surgical complications and readmissions decrease the quality measure and increase 90-day episode spending. Figure 10 presents the analogous spending gradient across markets binned on $\hat{\alpha}$.

The remaining puzzle is why health effects turn *negative* when we reset familiarity. The explanation is that existing referral relationships themselves encode meaningful quality infor-

mation. We examine this by comparing specialist quality across the depth of the referral relationship, measured as the number of patients a PCP has referred to a given specialist and binned to match the familiarity intervals in the structural model. Figure 11 shows that deeper referral relationships correspond to higher-quality specialists. In markets with the strongest familiarity-quality correlation (the "High" tercile in Figure 12), keeping familiarity produces health improvements while resetting it produces losses. This pattern is consistent with accumulated referral relationships serving as quality sorting when information frictions prevent full quality learning.

The preceding analysis shows that quality-targeted reallocation produces meaningful health gains when PCPs already weight quality, but the estimated weight is small in most markets. As a final thought experiment, we consider a simulation that raises the utility weight on quality to match the weight on distance, the dominant non-quality factor in referral patterns. We implement this by replacing each PCP's belief with the true specialist success rate, removing specialist fixed effects and accumulated familiarity, and setting α in each market so that one standard deviation of quality moves utility by as much as one standard deviation of distance. Because this strips away all drivers of referrals other than quality, distance, and capacity, including any unobserved demand factors captured by the fixed effects, we view the results as an idealized upper bound on the gains from quality-directed referrals rather than the predicted effect of any feasible intervention. Figure 13 summarizes the resulting reallocation and health effects. Roughly half of all referrals are reallocated, with average improvements of approximately 150 to 200 successes per 10,000 patients (a 17 to 22 percent reduction in negative outcomes) in both the myopic and forward-looking models and positive health effects in nearly every market. These gains are an order of magnitude larger than under full information at the estimated α, reflecting that the value of quality information alone is bounded by the small estimated quality weight. While large, these upper bound magnitudes are consistent with the within-market variation in specialist quality presented in Figure 2.

6 Discussion

This paper examines how PCPs learn about specialist quality and make referral decisions in the context of orthopedic surgery. Using 100% Medicare FFS claims data spanning 11 years,

we estimate a structural learning model to understand how PCPs respond to patient outcomes, and we use counterfactual simulations to quantify the effects of informational frictions in this market.

Employing both a design-based event study and a structural model, we find that PCPs modestly adjust their referrals based on negative patient outcomes. PCP referral patterns are heavily influenced by familiarity with specialists, and the estimated weight on quality is small relative to other referral determinants. Despite the small quality weight, counterfactual analyses indicate that full information about specialist quality could meaningfully reallocate referrals and improve patient outcomes.

When PCPs retain their accumulated familiarity with specialists, full information nudges referrals toward better specialists within established referral networks, producing modest health improvements. When familiarity is reset, however, the quality signal is too weak to direct the resulting reallocation, and health effects reverse. This asymmetry arises because existing referral relationships encode quality information. Indeed, we document empirically that PCP-specialist pairs with stronger patient histories are associated with higher specialist quality across markets, consistent with familiarity gradually sorting PCPs toward better specialists. Resetting familiarity destroys this accumulated sorting. With the quality weight small relative to the dispersion of other referral determinants (specialist fixed effects, distance preferences, and congestion), the resulting reallocation proceeds largely independently of quality, and markets where familiarity had encoded the most quality information experience the largest health losses.

From a policy perspective, our results suggest that quality transparency initiatives, such as public reporting of specialist performance, may be most effective when they complement rather than disrupt existing referral relationships. Information campaigns or decision-support tools could help PCPs identify better specialists within their existing networks, leveraging the familiarity channel rather than working against it. Conversely, interventions that reset referral patterns may require stronger quality signals to prevent the reallocation from being dominated by non-quality factors.

References

Ackerberg, Daniel A., "Advertising, Learning, and Consumer Choice in Experience Good Markets: An Empirical Examination," *International Economic Review*, 2003, *44* (3), 1007–1040.

Agha, Leila, Brigham Frandsen, and Jambes B. Rebitzer, "Causes and Consequences of Fragmented Care Delivery: Theory, Evidence, and Public Policy," Working Paper 23078, National Bureau of Economic Research April 2017.

_ , Keith Marzilli Ericson, Kimberley H. Geissler, and James B. Rebitzer, "Team Relationships and Performance: Evidence from Healthcare Referral Networks," *Management Science*, May 2022, *68* (5), 3735–3754.

Banerjee, Abhijit V. and Esther Duflo, "Reputation Effects and the Limits of Contracting: A Study of the Indian Software Industry," *The Quarterly Journal of Economics*, 2000, *115* (3), 989–1017.

Barkowski, Scott, "Physician Response to Prices of Other Physicians: Evidence from a Field Experiment," Working Paper, Clemson University 2018.

Barnett, Michael L, Bruce E Landon, A James O'malley, Nancy L Keating, and Nicholas A Christakis, "Mapping physician networks with self-reported and administrative data," *Health services research*, 2011, *46* (5), 1592–1609.

_ , Nancy L Keating, Nicholas A Christakis, A James O'Malley, and Bruce E Landon, "Reasons for choice of referral physician among primary care and specialist physicians," *Journal of general internal medicine*, 2012, *27* (5), 506–512.

Barnett, Michael L., Zirui Song, and Bruce E. Landon, "Trends in physician referrals in the United States, 1999-2009," *Archives of Internal Medicine*, January 2012, *172* (2), 163–170.

Bayer, Patrick and Christopher Timmins, "Estimating Equilibrium Models of Sorting Across Locations," *The Economic Journal*, March 2007, *117* (518), 353–374.

Berry, Steven, James Levinsohn, and Ariel Pakes, "Automobile prices in market equilibrium," *Econometrica: Journal of the Econometric Society*, 1995, pp. 841–890.

Brezzi, Monica and Tze Leung Lai, "Optimal learning and experimentation in bandit problems," *Journal of Economic Dynamics and Control*, 2002, *27* (1), 87–108.

Chan, Tat, Chakravarthi Narasimhan, and Ying Xie, "Treatment effectiveness and side effects: A model of physician learning," *Management Science*, 2013, *59* (6), 1309–1325.

Chernew, M., G. Gowrisankaran, and D.P. Scanlon, "Learning and the value of information: Evidence from health plan report cards," *Journal of Econometrics*, 2008, *144* (1), 156–174.

Chernew, Michael, Zack Cooper, Eugene Larsen Hallock, and Fiona Scott Morton, "Physician agency, consumerism, and the consumption of lower-limb MRI scans," *Journal of Health Economics*, March 2021, *76*, 102427.

Ching, Andrew T., "Consumer learning and heterogeneity: Dynamics of demand for prescription drugs after patent expiration," *International Journal of Industrial Organization*, November 2010, *28* (6), 619–638.

_ , Tülin Erdem, and Michael P. Keane, "Learning Models: An Assessment of Progress, Challenges, and New Developments," *Marketing Science*, November 2013, *32* (6), 913–938.

_ , _ , and _ , "Empirical Models of Learning Dynamics: A Survey of Recent Developments," in Berend Wierenga and Ralf van der Lans, eds., *Handbook of Marketing Decision Models*, International Series in Operations Research & Management Science, Cham: Springer International Publishing, 2017, pp. 223–257.

Coscelli, Andrea and Matthew Shum, "An empirical model of learning and patient spillovers in new drug entry," *Journal of Econometrics*, October 2004, *122* (2), 213–246.

Crawford, Gregory S and Matthew Shum, "Uncertainty and learning in pharmaceutical demand," *Econometrica*, 2005, *73* (4), 1137–1173.

Cutler, David, Leemore Dafny, David C. Grabowski, Steven Lee, and Christopher Ody, "Vertical Integration and Cream Skimming of Profitable Referrals: The Case of

Hospital-Owned Skilled Nursing Facilities," *American Economic Journal: Economic Policy*, May 2026, *18* (2), 1–33.

Dickstein, Michael, "Efficient provision of experience goods: Evidence from antidepressant choice," Working Paper 18-17, New York University, Leonard N. Stern School of Business 2018.

Dranove, David and Christopher Ody, "Employed for higher pay? How Medicare facility fees affect hospital employment of physicians," *American Economics Journal: Economic Policy*, 2019, *11* (4), 249–271.

_ and Ginger Zhe Jin, "Quality Disclosure and Certification: Theory and Practice," *Journal of Economic Literature*, 2010, *48* (4), 935–963.

Egan, Mark, Gregor Matvos, and Amit Seru, "The Market for Financial Adviser Misconduct," *Journal of Political Economy*, February 2019, *127* (1), 233–295.

Einav, Liran, Amy Finkelstein, Yunan Ji, and Neale Mahoney, "Randomized trial shows healthcare payment reform has equal-sized spillover effects on patients not targeted by reform," *Proceedings of the National Academy of Sciences*, August 2020, *117* (32), 18939–18947.

_ , _ , _ , and _ , "Voluntary Regulation: Evidence from Medicare Payment Reform*," *The Quarterly Journal of Economics*, February 2022, *137* (1), 565–618.

Erdem, Tülin and Michael P. Keane, "Decision-Making under Uncertainty: Capturing Dynamic Brand Choice Processes in Turbulent Consumer Goods Markets," *Marketing Science*, 1996, *15* (1), 1–20.

Fader, Peter S., James M. Lattin, and John D. C. Little, "Estimating Nonlinear Parameters in the Multinomial Logit Model," *Marketing Science*, November 1992, *11* (4), 372–385.

Ferreyra, Maria Marta and Grigory Kosenok, "Learning About New Products: An Empirical Study of Physicians' Behavior," *Economic Inquiry*, 2011, *49* (3), 876–898.

Freedman, Rachel A, Elena M Kouri, Dee W West, and Nancy L Keating, "Racial/ethnic differences in patients' selection of surgeons and hospitals for breast cancer surgery," *JAMA oncology*, 2015, *1* (2), 222–230.

Gaynor, Martin, Carol Propper, and Stephan Seiler, "Free to Choose? Reform, Choice, and Consideration Sets in the English National Health Service," *American Economic Review*, November 2016, *106* (11), 3521–57.

Gennaioli, Nicola, Andrei Shleifer, and Robert Vishny, "Money Doctors," *The Journal of Finance*, 2015, *70* (1), 91–114. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/jofi.12188.

Gittins, J. C. and D. M. Jones, "A dynamic allocation index for the discounted multiarmed bandit problem," *Biometrika*, December 1979, *66* (3), 561–565.

Gittins, John C, "Bandit processes and dynamic allocation indices," *Journal of the Royal Statistical Society: Series B (Methodological)*, 1979, *41* (2), 148–164.

Gong, Qing, "Physician Learning and Treatment Choices Evidence from Brain Aneurysms," Working Paper, University of North Carolina at Chapel Hill 2018.

Israel, Mark, "Services as Experience Goods: An Empirical Examination of Consumer Learning in Automobile Insurance," *American Economic Review*, December 2005, *95* (5), 1444–1463.

Kaur, Ramandeep, Jennifer N. Perloff, Christopher Tompkins, and Christine E. Bishop, "Hospital Postacute Care Referral Networks: Is Referral Concentration Associated with Medicare-Style Bundled Payments?," *Health Services Research*, 2017, *52* (6), 2079–2098.

Kolstad, Jonathan T. and Michael E. Chernew, "Quality and Consumer Decision Making in the Market for Health Insurance and Health Care Services," *Medical Care Research and Review*, February 2009, *66* (1_suppl), 28S–52S.

Levitt, Steven D. and Chad Syverson, "Market Distortions When Agents Are Better Informed: The Value of Information in Real Estate Transactions," *The Review of Economics and Statistics*, 2008, *90* (4), 599–611.

Lin, Haizhen, Ian M. McCarthy, Michael Richards, and Christopher Whaley, "Owning the Agent: Hospital Influence on Physician Behaviors," *Journal of Human Resources*, 2025.

Linnainmaa, Juhani T., Brian T. Melzer, and Alessandro Previtero, "The Misguided Beliefs of Financial Advisors," *The Journal of Finance*, 2021, *76* (2), 587–621. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/jofi.12995.

Macchiavello, Rocco and Ameet Morjaria, "The Value of Relationships: Evidence from a Supply Shock to Kenyan Rose Exports," *American Economic Review*, September 2015, *105* (9), 2911–2945.

Mehrotra, Ateev, Christopher B Forrest, and Caroline Y Lin, "Dropping the Baton: Specialty Referrals in the United States," *The Milbank Quarterly*, March 2011, *89* (1), 39–68.

Pham, Hoangmai H, Ann S O'Malley, Peter B Bach, Cynthia Saiontz-Martinez, and Deborah Schrag, "Primary care physicians' links to other physicians through Medicare patients: the scope of care coordination," *Annals of internal medicine*, 2009, *150* (4), 236–242.

Richards-Shubik, Seth, Mark S. Roberts, and Julie M. Donohue, "Measuring quality effects in equilibrium," *Journal of Health Economics*, May 2022, *83*, 102616.

Sarsons, Heather, "Interpreting Signals in the Labor Market: Evidence from Medical Referrals," Working Paper, Harvard University 2023.

Schneider, Eric C. and Arnold M. Epstein, "Influence of Cardiac-Surgery Performance Reports on Referral Practices and Access to Care — A Survey of Cardiovascular Specialists," *New England Journal of Medicine*, July 1996, *335* (4), 251–256. _eprint: https://www.nejm.org/doi/pdf/10.1056/NEJM199607253350406.

Stanton, Christopher T. and Catherine Thomas, "Landing the First Job: The Value of Intermediaries in Online Hiring," *The Review of Economic Studies*, April 2016, *83* (2), 810–854.

Wright, John R., Divya K. Madhusudhan, David C. Lawrence, Sharon A. Watts, Daniel J. Lord, Christopher Whaley, and Dena M. Bravata, "Costs of Specialist

Referrals From Employer-Sponsored Integrated Health Care Clinics Are Lower Than Those From Community Providers," *Journal of General Internal Medicine*, November 2022, *37* (15), 3861–3868.

Zeltzer, Dan, "Gender Homophily in Referral Networks: Consequences for the Medicare Physician Earnings Gap," *American Economic Journal: Applied Economics*, April 2020, *12* (2), 169–97.

Tables and Figures

Table 1: Descriptive Statistics for Patients and Referrals[a]

	2008-2012	2013-2018	Overall
Patient Characteristics			
Age	74.611	74.100	74.311
	(6.305)	(6.181)	(6.238)
White	0.943	0.931	0.936
	(0.231)	(0.253)	(0.244)
Black	0.037	0.037	0.037
	(0.189)	(0.190)	(0.189)
Male	0.353	0.373	0.365
	(0.478)	(0.484)	(0.481)
PCP Characteristics			
Practice Size	12.527	18.442	16.002
	(20.311)	(27.265)	(24.807)
Male	0.805	0.770	0.785
	(0.396)	(0.421)	(0.411)
Specialist Characteristics			
Practice Size	4.251	6.371	5.497
	(4.716)	(7.755)	(6.752)
Male	0.991	0.988	0.989
	(0.096)	(0.107)	(0.103)
Distance			
to PCP	30.949	28.481	29.494
	(160.244)	(159.272)	(159.676)
to Hospital	40.424	40.449	40.439
	(167.580)	(167.529)	(167.549)
90-day Readmission	0.081	0.066	0.072
	(0.272)	(0.249)	(0.259)
90-day Mortality	0.007	0.005	0.006
	(0.081)	(0.069)	(0.074)
90-day Complication	0.107	0.089	0.097
	(0.309)	(0.285)	(0.296)
Failure	0.109	0.091	0.099
	(0.312)	(0.288)	(0.298)
Observations	831,056	1,183,271	2,014,327

[a]Mean values with standard deviations in parentheses. As discussed in the main text, the period from 2008-2012 is our baseline period used to form histories of PCP/specialist pairs over a common time period, and the period from 2013-2018 is our estimation period. We present summary statistics separately by those time periods for consistency with our final estimation.

Table 2: Descriptive Statistics for PCPs and PCP/Specialist Pairs[a]

	2008-2012	2013-2018	Overall
Total Referrals (out)	3.809	4.398	4.134
	(2.940)	(3.478)	(3.261)
Network Size	2.666	3.109	2.910
	(1.580)	(1.831)	(1.737)
Total Failures	0.416	0.401	0.408
	(0.696)	(0.684)	(0.690)
Total Referrals (in)	28.685	30.044	29.468
	(28.165)	(30.520)	(29.552)
Referrals per Specialist	1.429	1.415	1.420
	(1.009)	(1.021)	(1.016)
Failure Rate	0.111	0.093	0.101
	(0.293)	(0.271)	(0.280)
Running Referrals		4.445	
		10.453	
Running Failure Rate		0.101	
		0.201	
Observations	581,641	836,440	1,418,081

[a]Mean values calculated per year, with standard deviations in parentheses. As discussed in the main text, the period from 2008-2012 is our baseline period used to form histories of PCP/specialist pairs over a common time period, and the period from 2013-2018 is our estimation period. We present summary statistics separately by those time periods for consistency with our final estimation.

Table 3: Re-Referral to the Same Specialist After a Failure[a]

	Effect	Std. error	Base	Relative (%)
Next referral to the same specialist				
All events	−0.0110	0.0011	0.160	−6.8
With PCP fixed effects	−0.0106	0.0011		
No prior referrals	−0.0126	0.0013	0.059	−21.2
1–4 prior referrals	−0.0100	0.0018	0.131	−7.6
5 or more prior referrals	−0.0093	0.0025	0.281	−3.3

[a]Estimates of β from Equation (1), with specialist-by-year fixed effects and indicators for the pair's number of prior referrals (top-coded at five). Base is the mean outcome following a success. Standard errors clustered by specialist and by PCP. The sample contains 1,190,521 events, of which 112,871 are failures.

Table 4: Structural Parameter Estimates, Myopic Model[a]

			Percentile				
Parameter	Mean	(SD/SE)	10th	25th	50th	75th	90th
α (utility weight on outcome)							
($\eta = 1$)	0.2020	(0.1791)	0.0000	0.0000	0.1166	0.2830	0.5372
($\eta = 5$)	0.3972	(0.3723)	0.0000	0.0000	0.2121	0.5605	1.0891
π (utility weight on distance)							
($\eta = 1$)	-0.0694	(0.0060)	-0.0990	-0.0824	-0.0678	-0.0525	-0.0384
($\eta = 5$)	-0.0695	(0.0061)	-0.1002	-0.0826	-0.0679	-0.0528	-0.0390
ρ (prior mean)							
(all η)	0.8996	(0.0117)	0.8853	0.8920	0.8995	0.9068	0.9139
γ (congestion effect, per 100 patients)							
($\eta = 1$)	-0.0913	(0.0153)					
($\eta = 5$)	-0.1158	(0.0169)					

κ_b (familiarity)	Range of e_{ijt}						
	1	2	3	4	5	[6,7]	[8,10]
($\eta = 1$)	1.1075	1.4307	1.6991	1.8802	2.0293	2.2027	2.3996
	(0.0982)	(0.1141)	(0.1278)	(0.1418)	(0.1574)	(0.1497)	(0.1712)
($\eta = 5$)	1.1143	1.4375	1.7084	1.8895	2.0381	2.2121	2.4063
	(0.0982)	(0.1141)	(0.1279)	(0.1420)	(0.1577)	(0.1503)	(0.1716)

[a]Estimates of specification in Equation (11), using the procedure described in Section 5.3, with different values of η as indicated. Parameters α, π, ρ are estimated separately by HRR; the table reports their distribution, where SD is standard deviation across HRRs. For γ, the "mean" is the point estimate and SE is the standard error from a single nationwide estimation. All familiarity coefficients are presented as the mean across HRRs with standard errors in parentheses. Results limited to HRRs in which our constrained maximization routine converged, 280 HRRs for $\eta = 1$ and 281 for $\eta = 5$.

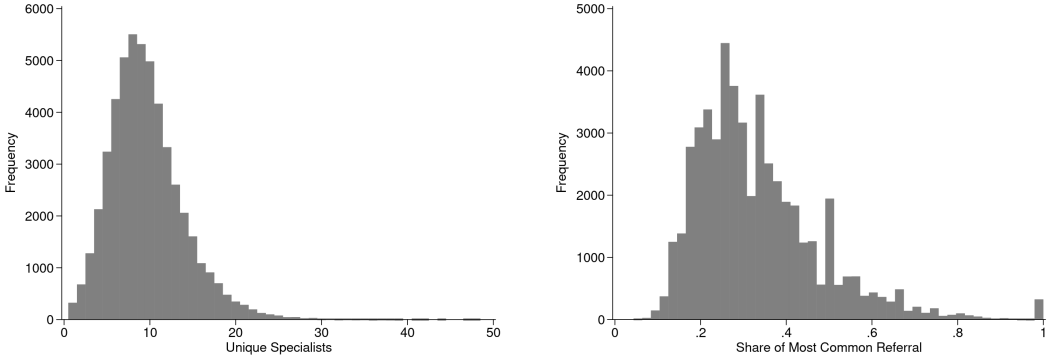
Table 5: Structural Parameter Estimates, Forward-looking Model[a]

Parameter	Mean	(SD/SE)	Percentile 10th	25th	50th	75th	90th
α (utility weight on outcome)							
($\eta = 1$)	0.2630	(0.2308)	0.0000	0.0000	0.1607	0.3652	0.7071
($\eta = 5$)	0.5050	(0.4936)	0.0000	0.0000	0.2581	0.7280	1.3807
π (utility weight on distance)							
($\eta = 1$)	-0.0692	(0.0060)	-0.0985	-0.0819	-0.0678	-0.0523	-0.0390
($\eta = 5$)	-0.0698	(0.0060)	-0.0998	-0.0829	-0.0685	-0.0531	-0.0384
ρ (prior mean)							
(all η)	0.8996	(0.0117)	0.8853	0.8920	0.8995	0.9068	0.9139
γ (congestion effect, per 100 patients)							
($\eta = 1$)	-0.0871	(0.0152)					
($\eta = 5$)	-0.1055	(0.0164)					

κ_b (familiarity)	Range of e_{ijt} 1	2	3	4	5	[6,7]	[8,10]
($\eta = 1$)	1.1559	1.4855	1.7583	1.9407	2.0900	2.2650	2.4631
	(0.1147)	(0.1329)	(0.1464)	(0.1604)	(0.1747)	(0.1684)	(0.1890)
($\eta = 5$)	1.2008	1.5391	1.8200	2.0045	2.1586	2.3331	2.5319
	(0.1449)	(0.1700)	(0.1861)	(0.2008)	(0.2150)	(0.2112)	(0.2315)

[a]Estimates of specification in Equation (12), using the procedure described in Section 5.3, with different values of η as indicated. Parameters α, π, ρ are estimated separately by HRR; the table reports their distribution, where SD is standard deviation across HRRs. For γ, the "mean" is the point estimate and SE is the standard error from a single nationwide estimation. All familiarity coefficients are presented as the mean across HRRs with standard errors in parentheses. Results limited to HRRs in which our constrained maximization routine converged, 277 HRRs for $\eta = 1$ and 279 for $\eta = 5$.

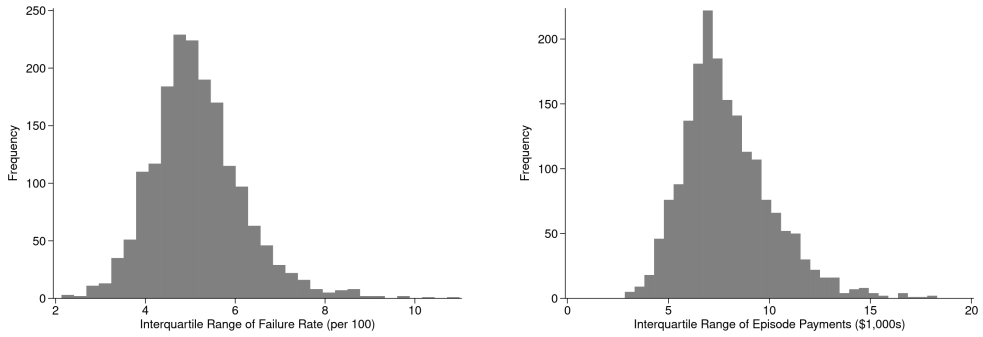
Figure 1: PCP Referral Networks over 2013-2018[a]



(a) Network Size (b) Highest Referral Share

[a]Panel (a) depicts the frequency histogram of PCP network size (number of unique specialists referred to) over the 2013-2018 estimation period. Panel (b) depicts the frequency histogram of the "highest-share" specialist per PCP over the same period.

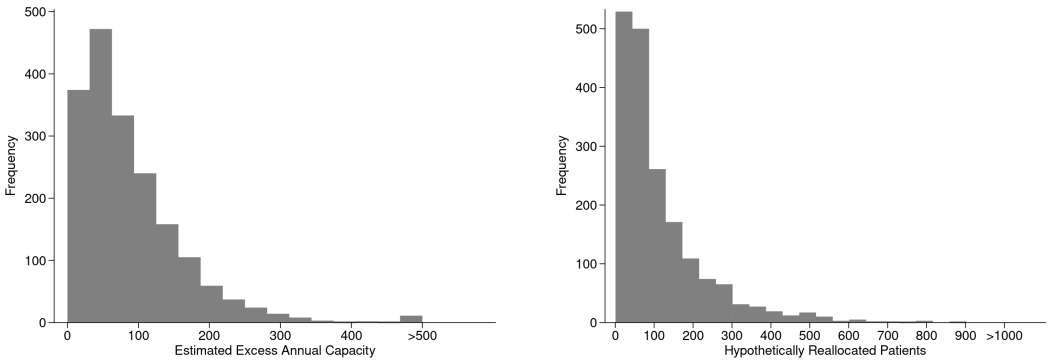
Figure 2: Within-Market Variation in Specialist Quality and Spending[a]



(a) Failure Rate (b) Episode Spending

[a]Panel (a) depicts the interquartile range of failure rates per 100 surgeries across specialists within each HRR over the 2013-2018 estimation period. Panel (b) depicts the interquartile range of mean episode spending per specialist per HRR over the same period. Episode spending is measured as the total Medicare payments over a 90-day period beginning from the date of admission.

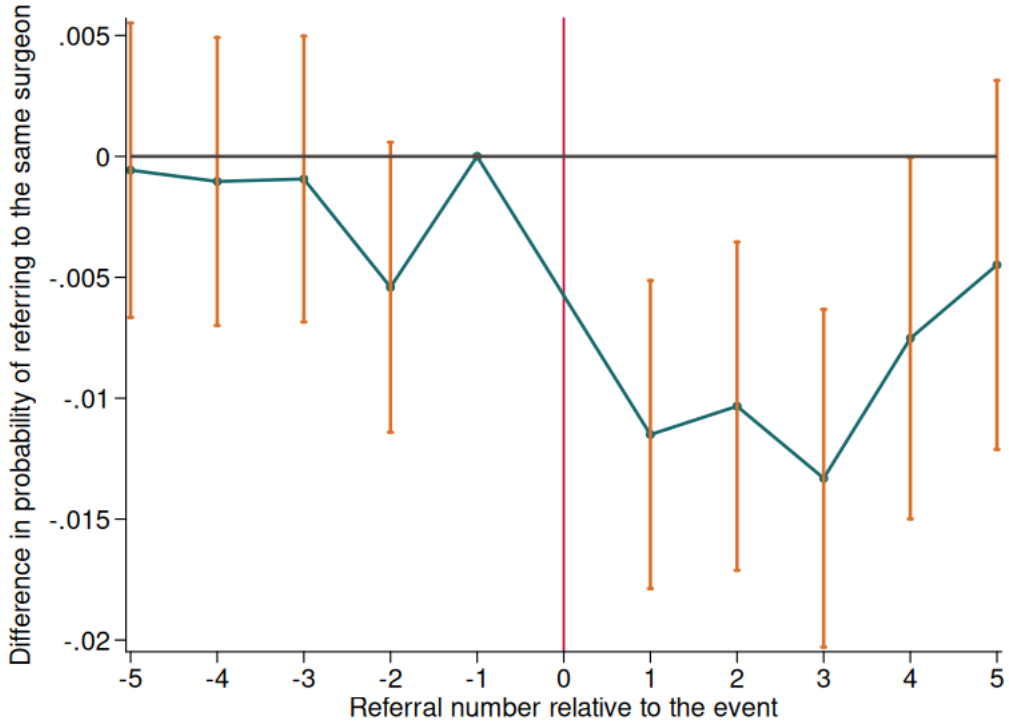
Figure 3: Specialist Capacity and Hypothetical Reallocation[a]



(a) Potential Excess Capacity (b) Hypothetical Reallocation

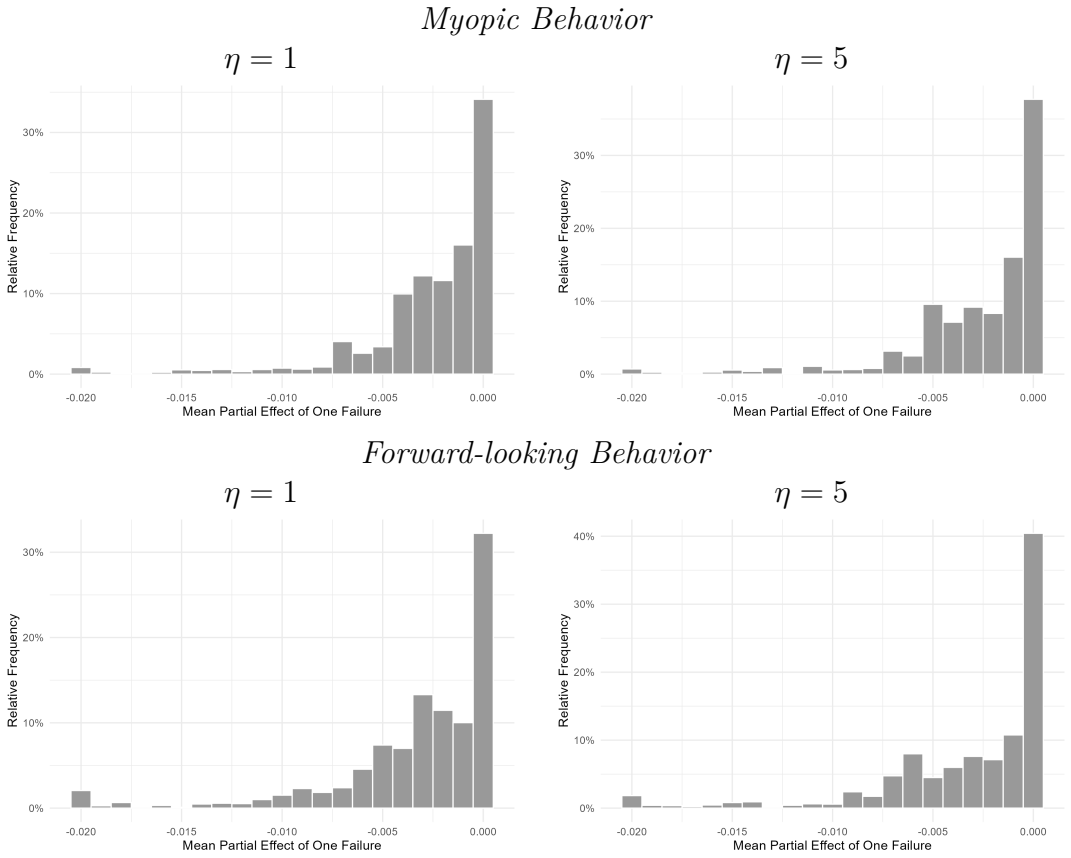
[a]Panel (a) depicts the estimated number of additional annual procedures that could be performed collectively among the specialists in the top 25 percent of the quality distribution. Excess capacity for a single specialist is measured as the difference between the 75th percentile of that specialist's count of yearly operations less the observed count of operations in that year. Panel (b) depicts the count of procedures among the lowest 25th percentile of specialists, aggregated across all specialists in a given HRR.

Figure 4: Re-Referral Event Study in Referral Order[a]



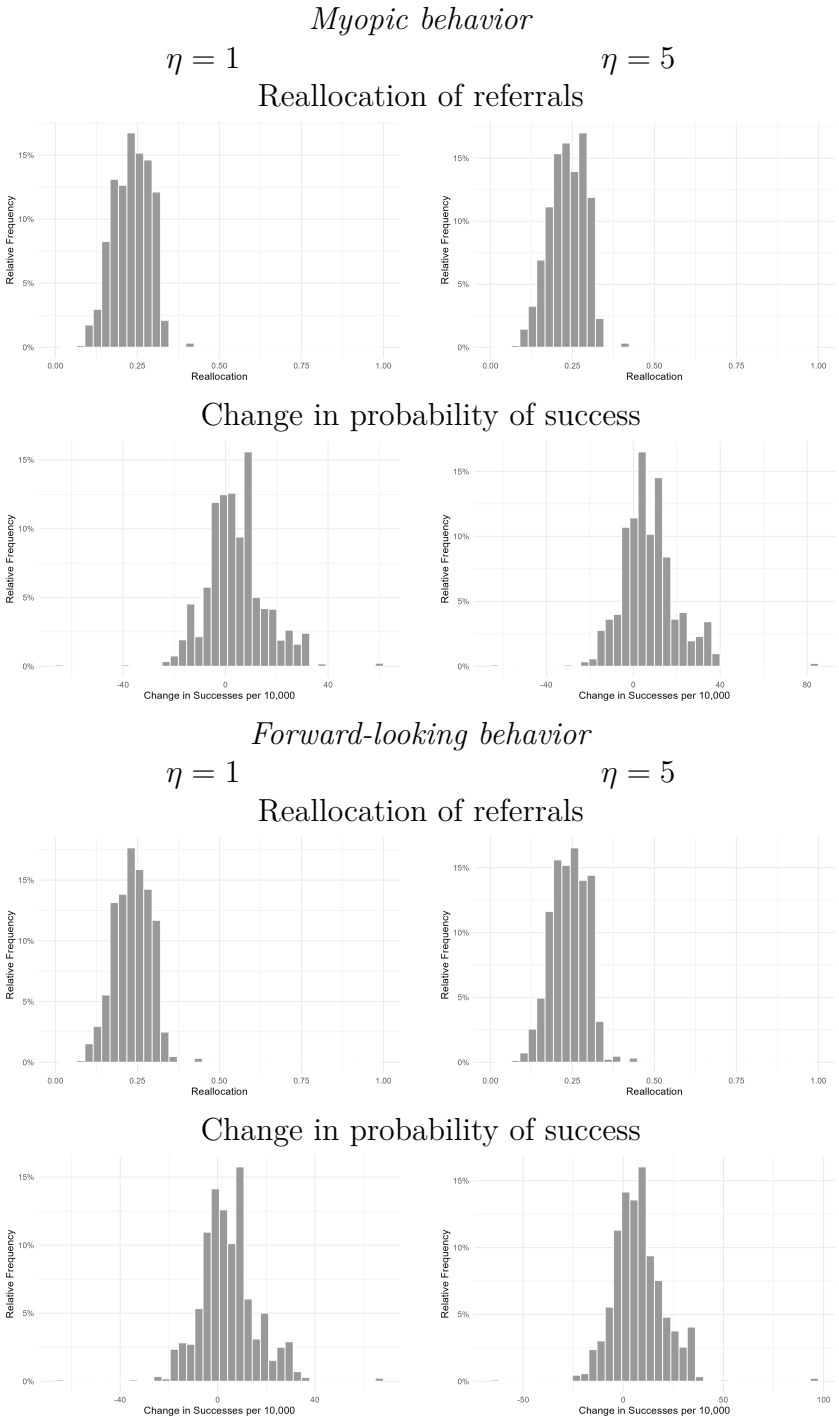
[a]The figure plots estimates of δ_k from Equation (2), the difference between failure and success windows in the probability that PCP i refers patient k to specialist j. The referral at $k = 0$ defines the window, and a failure at $k = 0$ is the treatment. Each window includes the PCP's five referrals before and after the referral at $k = 0$. We keep a window only if none of the PCP's other referrals in it resulted in a failure. The specification includes window fixed effects, specialist-by-year fixed effects interacted with event time, and PCP fixed effects interacted with event time. The coefficient at $k = -1$ is normalized to 0. Bars show 95% confidence intervals with standard errors clustered by specialist and by PCP.

Figure 5: Partial Effect of Failures on Referral Probabilities[a]



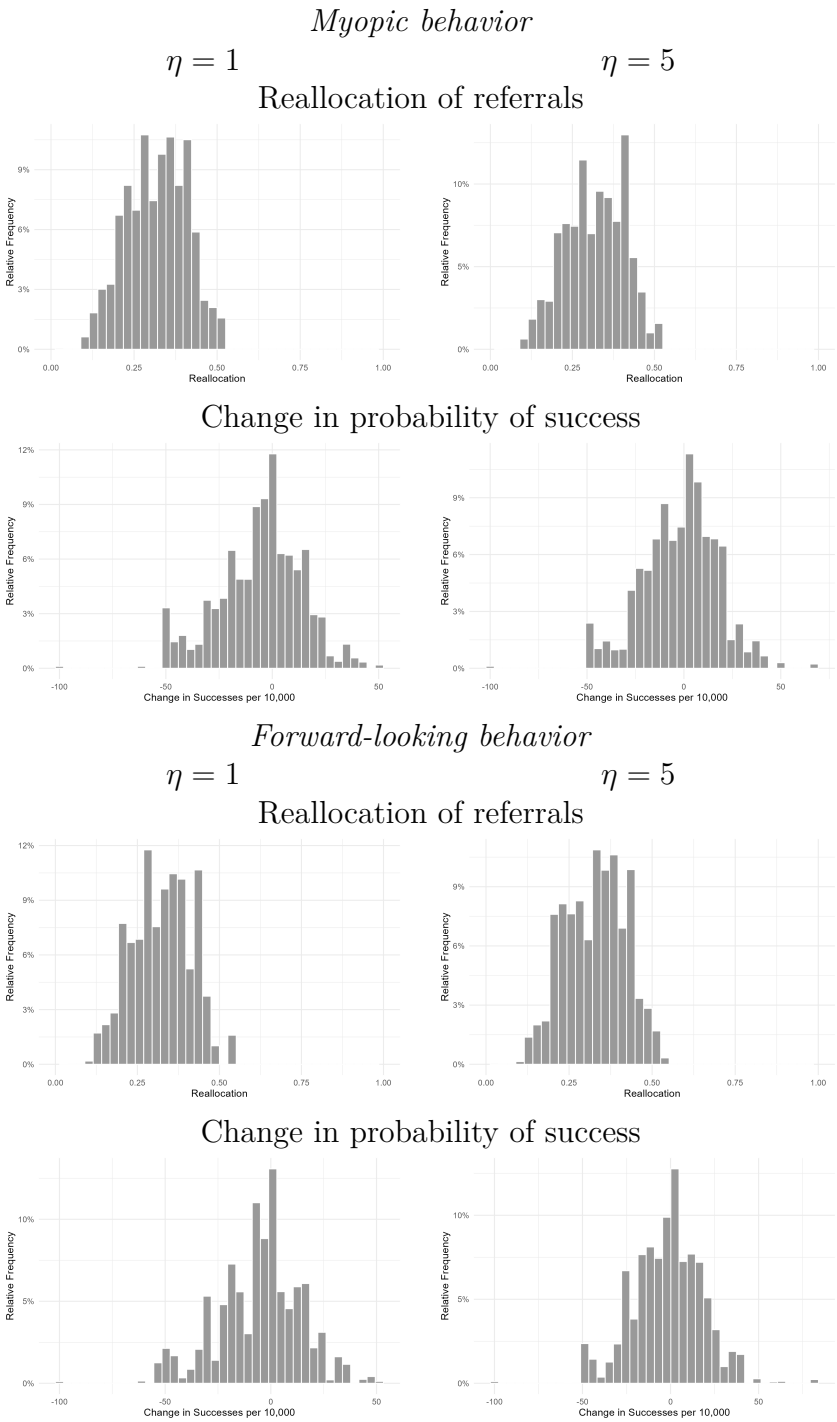
[a]Figure depicts the mean change in referral probabilities among PCP-specialist pairs with some referral history, averaged within an HRR and presented as a histogram across HRRs. Results limited to HRRs in which our constrained maximization routine converged, 280 (277) HRRs in the myopic (forward-looking) case for $\eta = 1$ and 281 (280) in the myopic (forward-looking) case for $\eta = 5$.

Figure 6: Counterfactual Changes with Full Information, Existing Familiarity[a]



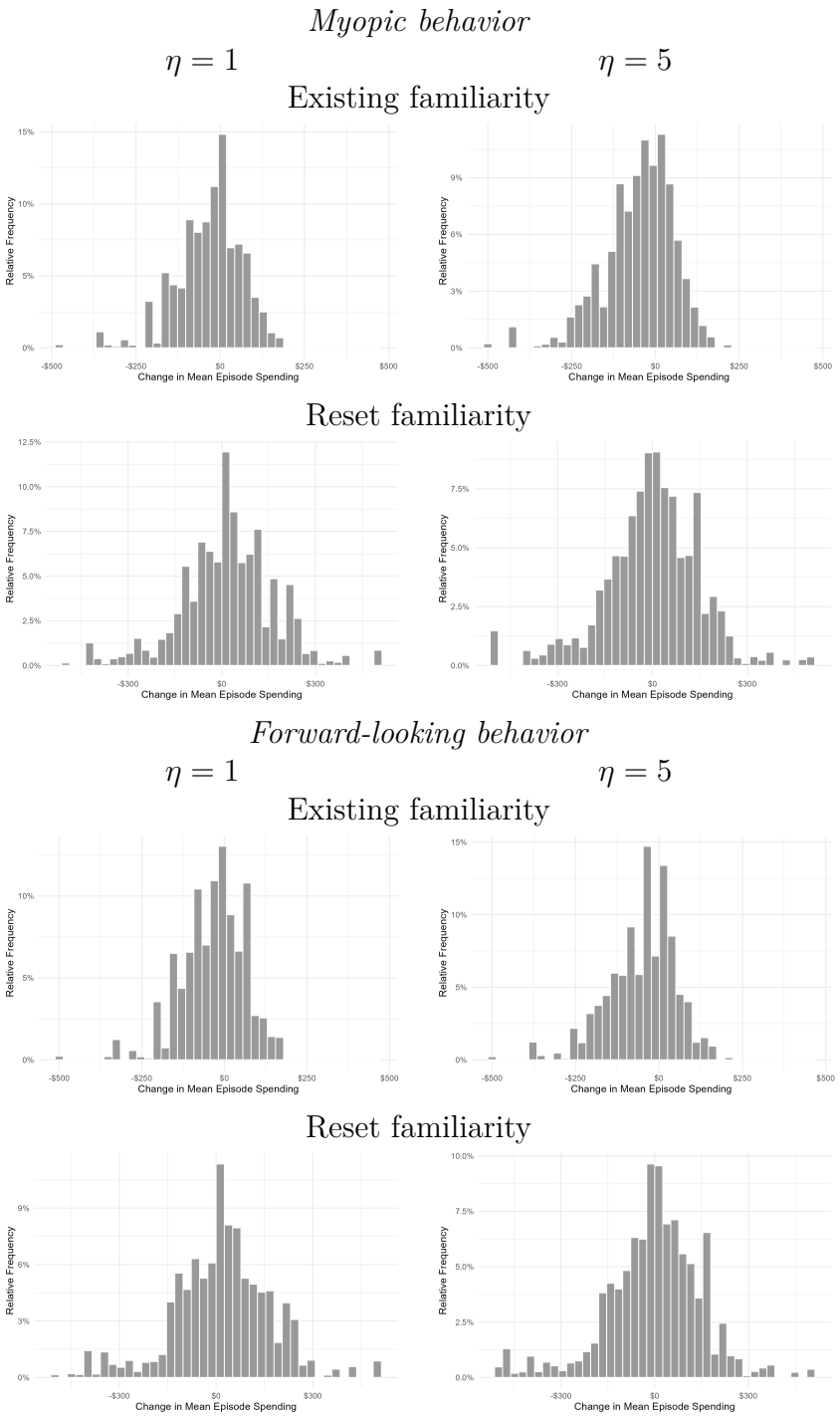
[a]Figure depicts the share of referrals reallocated and the resulting change in patient outcomes under a counterfactual in which PCPs have full information about specialist quality and in which familiarity is initialized to observed values as of January 2013. Estimates are averaged across individuals within an HRR and presented as a histogram across HRRs. Results are limited to HRRs in which our constrained maximization routine converged, with 280 HRRs (myopic, $\eta = 1$), 281 HRRs (myopic, $\eta = 5$), 277 HRRs (forward-looking, $\eta = 1$), and 280 HRRs (forward-looking, $\eta = 5$).

Figure 7: Counterfactual Changes with Full Information, Reset Familiarity[a]



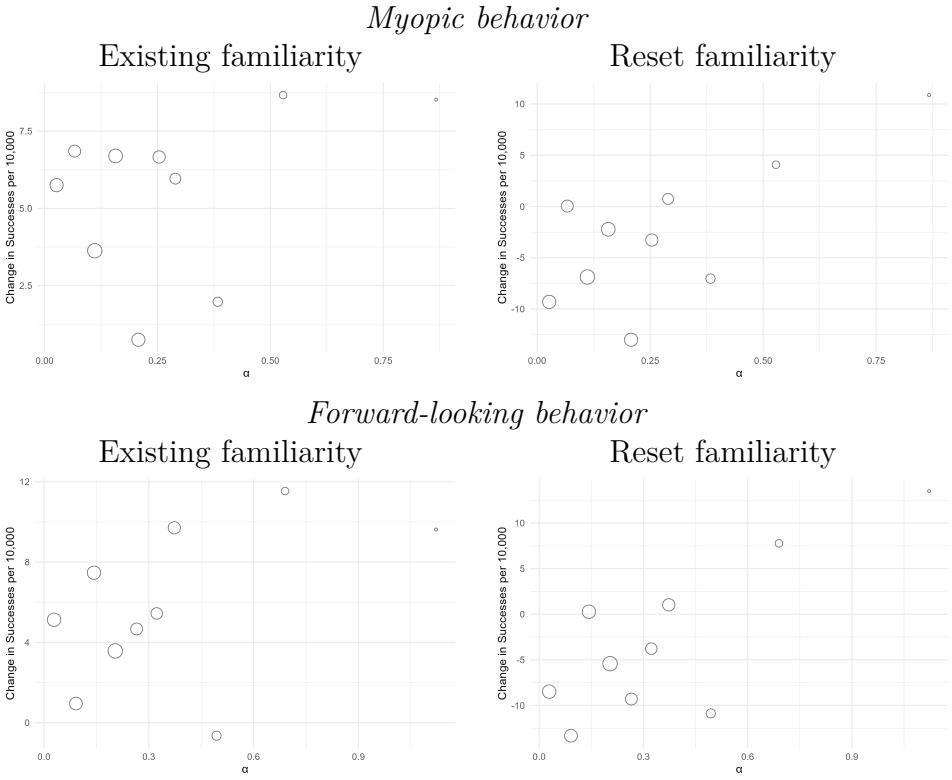
[a]Figure depicts the share of referrals reallocated and the resulting change in patient outcomes under a counterfactual in which PCPs have full information about specialist quality and in which familiarity is initialized to 0. Estimates are averaged across individuals within an HRR and presented as a histogram across HRRs. Results are limited to HRRs in which our constrained maximization routine converged, with 280 HRRs (myopic, $\eta = 1$), 281 HRRs (myopic, $\eta = 5$), 277 HRRs (forward-looking, $\eta = 1$), and 280 HRRs (forward-looking, $\eta = 5$).

Figure 8: Counterfactual Changes in Episode Spending with Full Information[a]



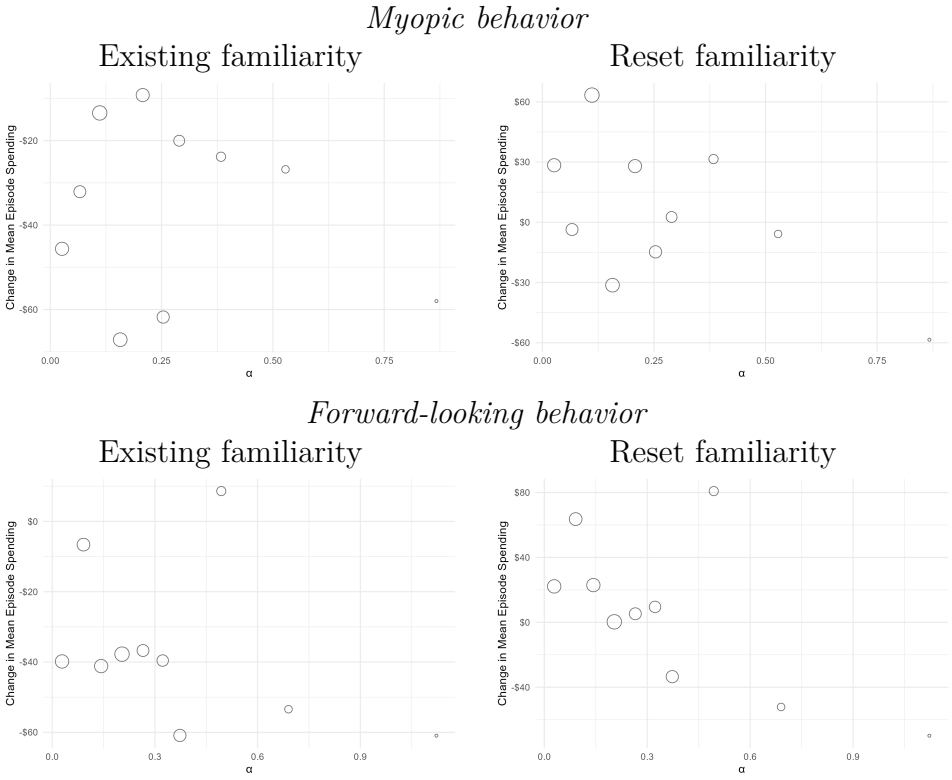
[a]Figure depicts the change in 90-day episode spending under each full-information counterfactual, averaged across individuals within an HRR and presented as a histogram across HRRs. The "existing familiarity" counterfactual initializes familiarity at observed values as of January 2013; the "reset familiarity" counterfactual initializes familiarity to 0.

Figure 9: Health Effects of Full Information by Estimated Learning Parameter $(\eta = 1)^a$



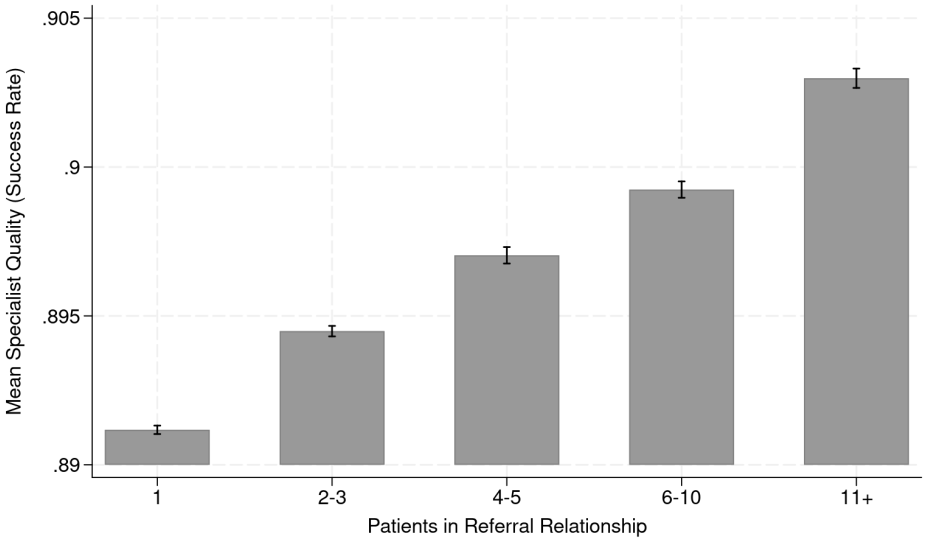
[a]Figure depicts the mean change in successes per 10,000 patients under each counterfactual, binned into ten patient-weighted quantiles of the estimated learning parameter $\hat{\alpha}$ within each HRR ($\hat{\alpha} > 0.001$). Marker size reflects the total patients in the bin. Corresponding figures for $\eta = 5$ appear in the supplemental appendix.

Figure 10: Episode Spending Effects of Full Information by Estimated Learning Parameter ($\eta = 1$)[a]



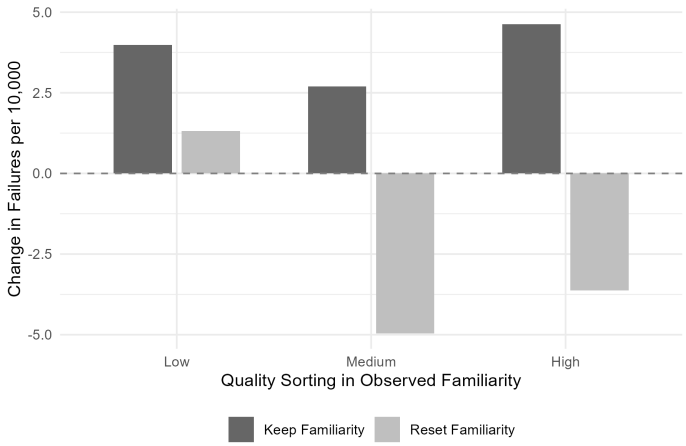
[a]Figure depicts the mean change in episode spending under each counterfactual, binned into ten patient-weighted quantiles of the estimated learning parameter $\hat{\alpha}$ within each HRR ($\hat{\alpha} > 0.001$). Marker size reflects the total patients in the bin. Corresponding figures for $\eta = 5$ appear in the supplemental appendix.

Figure 11: Familiarity and Specialist Quality[a]



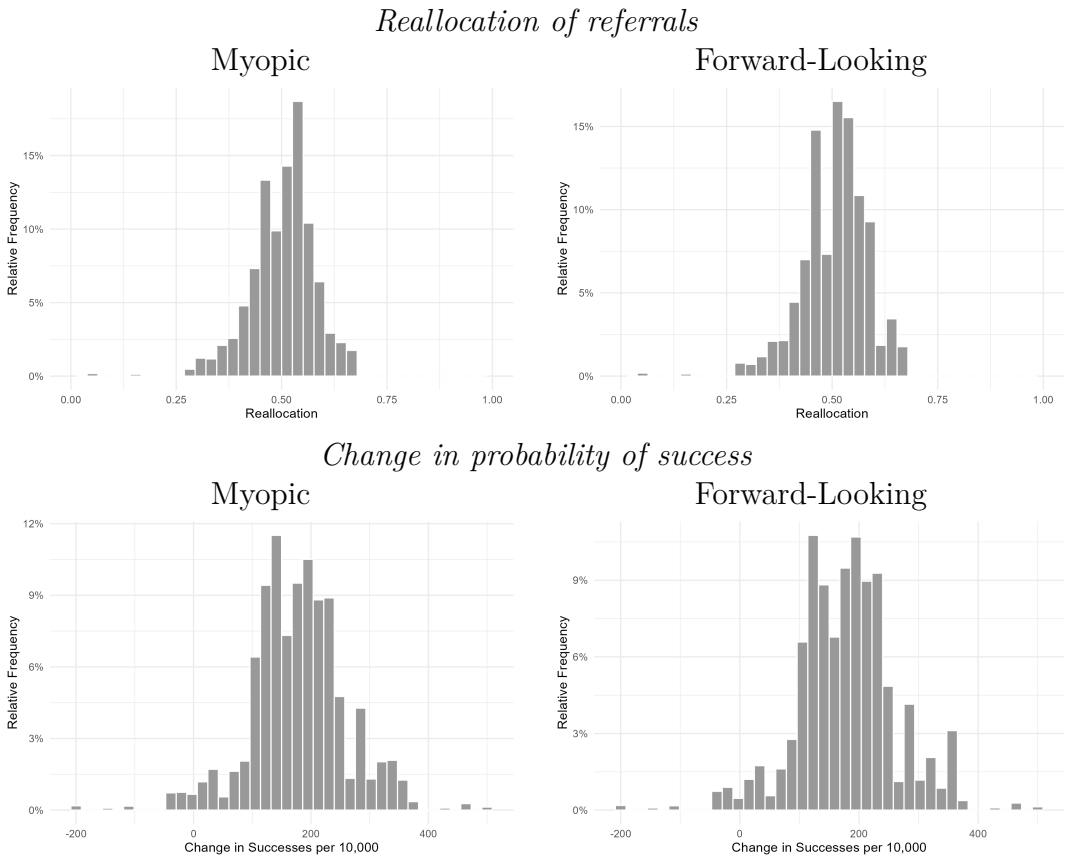
[a]Figure depicts mean specialist quality (success rate) by the number of patients a PCP has referred to a given specialist. Patients are grouped into bins: 1, 2–3, 4–5, 6–10, and 11+. Error bars represent 95% confidence intervals.

Figure 12: Health Effects of Full Information by Quality Sorting in Familiarity[a]



[a]Figure depicts the mean change in successes per 10,000 patients under full information, separately for the "existing familiarity" and "resetting familiarity" counterfactuals. Markets are grouped into terciles based on the within-market correlation between referral relationship duration and specialist quality. Higher terciles indicate markets where longer referral relationships are more strongly associated with higher specialist quality.

Figure 13: Counterfactual Changes with Quality-Maximizing Referrals ($\eta = 1$)[a]



[a]Figure depicts the share of referrals reallocated and the resulting change in probability of success under a counterfactual in which each PCP's belief is replaced with the true specialist success rate, specialist fixed effects and accumulated familiarity are set to zero, and α within each HRR is calibrated to match the larger of the distance or congestion utility components. Results are averaged across individuals within an HRR and presented as a histogram across HRRs. Corresponding figures for $\eta = 5$ appear in the supplemental appendix.